Restoring America's Nuclear Fuel Reprocessing Life Cycle

A Policy White Paper

A Case Study in How and Why Moving Spent Nuclear Fuel from the Liability to the Asset Column Can Lower Costs and Increase Profits for Existing and New Reactor Builds While Reducing Long-Term Environmental Risks

Author; Keith J Maxwell
October 2026

Export Control Statement

Notice: *This white paper contains only publicly available, policy-level information drawn from government reports, official statements, company announcements, and published policy studies. It deliberately excludes technical data, process information, flowsheets, chemistry, equipment, or design details that could be subject to U.S. export controls, including 10 CFR Part 810 (DOE) [1], 10 CFR Part 110 (NRC) [2], the International Traffic in Arms Regulations (ITAR), and the Export Administration Regulations (EAR). It does not discuss enrichment, weapons-related information, or safeguards vulnerabilities. It is not intended to provide, and does not provide, assistance in the production of special nuclear material. Technical topics are described only at the conceptual level found in public encyclopedias and government policy summaries.*

Readers seeking technical information on fuel-cycle technologies should consult the relevant licensing authorities and comply with all applicable export control regulations.

Contents

Executive Summary

1. The Problem: A Growing Liability with No Exit
2. Technical Benefits (Conceptual Overview)
3. The Financial and Accounting Case
4. Reducing Long-Term Environmental Risk
5. Strengthening Nonproliferation Through Recycling and MOX
6. Reducing Long-Term Federal Expenditures
7. The Historical Record
8. International Comparison

9. Policy and Political Changes Needed
10. Statutory, Regulatory, and Policy Changes Required
11. Building a Global Closed Fuel Cycle Cooperative: International Policy, Agreements, and Export Controls
12. An Advocacy Plan: Nonproliferation and Roadblock Removal
13. Conclusions and Recommendations
About the Author
References

Executive Summary

For chief financial officers, spent nuclear fuel is today a cost with no end date: an on-site storage obligation embedded in asset retirement estimates and offset only by slow litigation recoveries from a federal government that missed its 1998 acceptance date. A credible U.S. recycling pathway could change that. Spent fuel with a contracted destination becomes a finite, priceable obligation, and recovered material could, once contracts or markets support it, be recognized as salvage value or inventory rather than carried at zero. The potential effects are directional but meaningful: smaller and more predictable back-end obligations, stronger balance sheets and credit metrics, and lower risk premiums charged by lenders and investors for back-end uncertainty. For new small modular reactor (SMR) and advanced reactor projects, a defined back end and a secure domestic recycled-fuel supply could improve financeability, broaden investor appetite, and lower the cost of capital, a major driver of levelized cost for capital-intensive nuclear plants. The same program can reduce long-term environmental risks by shrinking the volume and duration of the waste that must be isolated, and it can reduce long-term federal liabilities.

The United States holds the largest inventory of commercial spent nuclear fuel in the world and has no operating pathway to dispose of it, recycle it, or derive value from it. As of December 31, 2024, U.S. reactor and storage sites held 95,117 metric tons of heavy metal (MTHM) of commercial spent fuel, a figure that grows by roughly 2,000 metric tons per year [3, 4]. Under the Nuclear Waste Policy Act of 1982 (NWPA) and the Standard Contract, the federal government promised to begin accepting that fuel by January 31, 1998 [5]. It has not. Taxpayers have paid about $12.2 billion from the Treasury Judgment Fund to utilities for the government's partial breach of contract through September 30, 2025, and the government's estimated remaining liability is $38.6 billion to $44.3 billion [6]. Meanwhile, the Nuclear Waste Fund, built from ratepayer fees, stands at roughly $51.4 billion and remains largely unusable for its intended purpose [7].

For utilities and investors, this policy stalemate has a concrete financial signature. Spent fuel is today carried almost entirely as a cost: an on-site storage burden folded into asset retirement obligations and decommissioning estimates, offset only in part by litigation recoveries from the federal government. The material itself, which still contains the large majority of its original energy potential [8], is assigned essentially no value. That treatment is rational under current law and markets, because there is no licensed U.S. pathway to recover and reuse the material. It is not an immutable fact of nuclear technology.

This paper makes the case that the United States should resume a closed, or partially closed, nuclear fuel cycle, deliberately designed for nonproliferation and economic discipline, so that recoverable materials in spent fuel can be recognized, at least in part, as an asset rather than solely as a liability. France has operated such a system for decades: Orano's La Hague complex has a licensed capacity of 1,700 tonnes of used fuel per year, and recycled mixed-oxide (MOX) fuel supplies about 10 percent of French nuclear generation [9]. French law separately classifies radioactive "materials" with a planned or envisaged use apart from "waste" [10], while EDF still books substantial provisions for back-end obligations. That dual treatment, with value

recognized where use is credible and liabilities booked where it is not, is the model this paper recommends adapting to U.S. accounting and regulatory practice.

The policy moment is unusually favorable. Executive Orders 14299 and 14302 (May 2025) directed DOE to recommend a national spent fuel and recycling policy and to authorize privately funded recycling and fuel-fabrication facilities on federal sites [11, 12]. In April 2026, DOE issued two Requests for Applications for private-sector recycling, including a commercial-scale demonstration at the Idaho site [13, 14]. In June 2026, the NRC proposed a rule that would allow reprocessing facilities to be licensed through a one-step 10 CFR Part 70 process [15]. Private firms have announced commercial plans, including Oklo's proposed recycling facility in Tennessee [16]. Congress held hearings on spent fuel policy in both chambers in September 2026 [17, 18].

This paper is candid about the obstacles. Independent studies, including MIT's 2011 fuel-cycle study and Harvard's 2003 economic analysis, concluded that under the market conditions of their time, once-through disposal was cheaper than recycling [19, 20]. Nonproliferation concerns ended U.S. commercial reprocessing in 1977 and remain legitimate. Recycling reduces but does not eliminate the need for a geologic repository. The recommendations that follow are built around those facts, not despite them.

Key recommendations

1. Amend the NWPA and the Standard Contract to create a voluntary "recycling pathway" under which spent fuel delivered to a licensed U.S. recycler satisfies, in whole or part, DOE's acceptance obligation, with a defined title, credit, and residual-waste disposal framework.
2. Restore disciplined access to the Nuclear Waste Fund for integrated back-end programs, including disposal of recycling residues, under the funding reforms recommended by GAO, the Blue-Ribbon Commission, and the Bipartisan Policy Center [21, 22, 23].
3. Ask FERC, state commissions, and the accounting standard-setting community to clarify how recovered nuclear materials may be recognized (for example, as net salvage value under the existing FERC Uniform System of Accounts, or as inventory under U.S. GAAP) once a contractual or market basis exists [24].
4. Finalize a modern NRC licensing framework for recycling facilities that incorporate safeguards-by-design and security requirements from the start [15, 25].
5. Adopt proliferation-resistant design principles as national policy for any federally supported recycling: no separated pure plutonium streams, co-located processing and fuel fabrication where practical, and eligibility for IAEA safeguards.
6. Treat environmental performance as a program requirement: a programmatic environmental impact statement comparing recycling with continued storage and direct disposal, with conditional benefits demonstrated rather than assumed (Section 4).
7. Make nonproliferation a design principle: balanced separation and use with no stockpiling of separated plutonium, Category I security, IAEA safeguards eligibility, and U.S. leadership on international norms (Section 5).

8. Use a recycling pathway to cap and reduce federal back-end exposure, including Judgment Fund damages, repository scale, and interim storage costs (Section 6).
9. Enact the specific statutory, regulatory, and executive-policy changes inventoried in Section 10. Most NRC rules need no change, and no nonproliferation protection should be weakened.
10. Build a Global Closed Fuel Cycle Cooperative that operates fully within U.S. nuclear export controls (10 CFR Parts 810 and 110, the EAR, AEA Sec. 123) and the NSG Guidelines, with updated authorizations and agreements for allied and supplier partners (Section 11).
11. Execute a phased pathway, from pilot demonstration on a federal site, to a first commercial facility, to fleet integration with fast reactors, with public milestones and independent review at each gate.

1. The Problem: A Growing Liability with No Exit

1.1 The inventory

Commercial spent fuel is stored at more than 70 sites in 35 states [4]. According to DOE's most recent inventory report, the commercial inventory reached 95,117 MTHM at the end of 2024, comprising 329,237 light-water reactor assemblies; about 53,300 MTHM was in dry storage and about 41,100 MTHM remained in pools [3]. DOE projects that the current fleet alone produce up to roughly 180,000 metric tons over its lifetime [3]. Every license extension, uprate, and restart, all of which the nation now wants, adds to that total.

1.2 The legal framework: the NWPA and the Standard Contract

The NWPA made the federal government responsible for permanent disposal of commercial spent fuel and high-level waste. In exchange, utilities signed the Standard Contract (10 CFR Part 961), paying a fee of 1.0 mill per kilowatt-hour (one-tenth of a cent) on nuclear generation into the Nuclear Waste Fund, plus a one-time fee for fuel used before April 1983 [5, 7]. DOE committed to begin accepting spent fuel by January 31, 1998. The 1987 NWPA amendments designated Yucca Mountain, Nevada, as the sole site to be characterized for the first repository.

1.3 Yucca Mountain stalled

DOE submitted the Yucca Mountain license application to the NRC in 2008. The program was defunded and DOE sought to withdraw the application in 2010; the NRC suspended its review, and policymakers have been at an impasse since [21]. After a 2013 court order, the NRC completed its safety evaluation using previously appropriated funds, but Congress has not appropriated funds to complete licensing, and there is no politically durable consensus on the site [verify: current NRC docket status]. GAO concluded in 2021 that congressional action is needed to break the impasse, including restructuring the Nuclear Waste Fund [21]; as of February 2026, GAO reported Congress had not acted on that recommendation [21].

1.4 The Nuclear Waste Fund

Because DOE could no longer make a legally adequate fee assessment, the U.S. Court of Appeals for the D.C. Circuit in 2013 ordered the Secretary of Energy to propose reducing the fee to zero, and collection stopped effective May 16, 2014 [26, 27]. Before suspension the fee generated roughly $750 million per year [27]. DOE reports that, through FY2025, the Fund received about $21.8 billion from commercial generators and $37.3 billion in interest and other income, and spent about $11.5 billion, leaving a current balance of about $51.4 billion as of September 30, 2025 [7]. DOE's summary also shows about $2.8 billion in receivables from commercial generators, largely associated with deferred one-time fees for pre-1983 fuel [7]. Under federal budget rules the Fund's balance cannot simply be spent; appropriations are scored against discretionary caps, a structural problem identified by the Blue-Ribbon Commission and the Bipartisan Policy Center [22, 23].

1.5 Judgment Fund damages

Utilities have sued, successfully, for the costs of storing fuel that the government was obligated to remove. Those payments come from the Treasury's permanent, indefinite Judgment Fund, not from DOE's budget or the Nuclear Waste Fund [21]. According to the FY2025 Financial Report of the U.S. Government, the estimated total liability is $50.8 billion to $56.5 billion; after the $12.2 billion paid through September 30, 2025, the remaining liability is $38.6 billion to $44.3 billion [6]. The Nuclear Waste Strategy Coalition has testified that this taxpayer liability grows by approximately $2 million per day [17].

Table 1. Key U.S. spent fuel figures (verified, with sources)

Metric	Value	As of	Source
Commercial spent fuel inventory	95,117 MTHM (329,237 assemblies)	Dec. 31, 2024	[3]
Annual discharge	~2,000 metric tons/yr	Current	[4]
Storage sites	70+ sites in 35 states	Current	[4]
Nuclear Waste Fund balance	~$51.4 billion	Sept. 30, 2025	[7]
NWF fee	Suspended (0.0 mill/kWh)	Since May 16, 2014	[26, 27]
Judgment Fund payments to date	~$12.2 billion cumulative	Sept. 30, 2025	[6]
Remaining federal liability (estimate)	$38.6-$44.3 billion	Sept. 30, 2025	[6]
Contractual acceptance date missed	Jan. 31, 1998	--	[5]

1.6 Why this is a financial & ecological balance-sheet problem

For an operating utility, every discharged assembly creates a long-tailed obligation: pool and dry-cask storage, security, insurance, regulatory fees, and eventual transfer costs. For a shutdown plant, spent fuel can be the last thing holding a site open long after decommissioning is otherwise complete, the so-called "stranded fuel" problem. Litigation recoveries offset part of the cost, but they arrive years after expenditure, require continuous legal effort, and depend on the government's ongoing breach. None of this produces revenue. It is a pure cost center that investors price into nuclear assets and new-build decisions.

2. Technical Benefits (Conceptual Overview)

Scope note: *This section describes benefits only at the conceptual level found in public encyclopedias and government policy summaries. It intentionally contains no process descriptions, chemistry, parameters, or design information.*

2.1 Energy remaining in spent fuel

Light-water reactor fuel is removed from service long before its potential energy is exhausted. ARPA-E states that more than 90 percent of the potential energy in U.S. used fuel remains [8]. Spent fuel consists mostly of uranium, with smaller fractions of plutonium, other actinides, and fission products [28]. In a recycling system, reusable materials are recovered and fabricated into new fuel, while the remainder is conditioned for disposal. France reports that recycled MOX fuel accounts for about 10 percent of its nuclear electricity [9].

2.2 Waste volume and radiotoxicity

Public assessments by DOE, ARPA-E, and the OECD Nuclear Energy Agency describe recycling as a way to reduce the volume, long-term heat load, and long-lived radiotoxicity of waste requiring geologic disposal, particularly when combined with reactors able to consume a broader range of actinides [8, 29]. The magnitude of the benefit depends on the fuel cycle chosen single recycle in light-water reactors offers modest gains, while multi-recycle in fast-spectrum reactors offer larger long-term gains [29]. Recycling does not eliminate the need for a repository; it changes what goes into it.

2.3 Repository space and design flexibility

Repository capacity is governed largely by heat, not by volume alone. Removing long-lived, heat-generating constituents from the disposal stream can, in principle, increase the effective capacity of a given repository and widen the range of acceptable geologies [29]. For U.S. policy, that matters because the statutory capacity limit and siting politics of a first repository have proven extraordinarily difficult.

2.4 Fuel security

Recovered uranium and other fissile material represent a domestic fuel resource already mined, enriched, and paid for. At a time when the United States is working to reduce reliance on foreign conversion and enrichment services, as reflected in Executive Order 14302 [11], a domestic recycled-fuel stream adds a strategic hedge, particularly for advanced reactors that need fuel forms not currently produced at scale.

2.5 Synergy with advanced and fast reactors

Several advanced reactor developers plan fast-spectrum designs that are well suited to recycled fuel. ARPA-E's CURIE program explicitly targets recovering materials from today's light-water reactor fuel as feedstock for advanced reactors, with goals for low fuel and disposal costs per kilowatt-hour [8]. Oklo has announced plans to recycle used fuel into fuel for its fast reactors [16]. A recycling capability and a fast-reactor fleet are mutually reinforcing each improves the economics of the other.

3. The Financial and Accounting Case

3.1 How spent fuel and back-end obligations are treated today

U.S. investor-owned utilities follow U.S. GAAP and, where rate-regulated, FERC's Uniform System of Accounts and state commission rules. Under current practice, the back end generally appears in several places:

- **Asset retirement obligations (AROs).** Under FASB ASC 410-20 (originally SFAS No. 143), utilities record the present value of legally required decommissioning costs. Decommissioning cost estimates for shutdown sites commonly include the costs of managing spent fuel on site until DOE removes it, although the NRC treats spent fuel management funding separately from radiological decommissioning funding [verify company-specific practice varies].
- **DOE contract recoveries.** Utilities typically recognize receivables for amounts recoverable from DOE under settlement agreements or judgments once recovery is probable, reducing the net cost of storage [verify practice varies by company].
- **Pre-1983 one-time fee obligations.** Some utilities that elected to defer the one-time fee still carry a spent nuclear fuel obligation to DOE on their balance sheets; DOE's NWF summary reports about $2.8 billion in receivables from commercial generators as of September 30, 2025 [7].
- **Nuclear fuel accounts.** FERC's Uniform System of Accounts tracks nuclear fuel through Accounts 120.1 to 120.5, with amortization charged to Account 518 (nuclear fuel expense) [24].

On the federal side, the government carries the Standard Contract breach as a contingent liability in the Financial Report of the U.S. Government, separate from the Nuclear Waste Fund's own financial statements [6, 30].

3.2 A provision already in the books

The FERC Uniform System of Accounts already contains the conceptual hook this paper seeks to reactivate. Account 120.4 describes spent fuel as assemblies removed from a reactor "pending reprocessing," and Account 120.5 defines the net cost of fuel assemblies subject to amortization as original cost "plus or less the expected net salvage value of uranium, plutonium, and other byproducts," with recovered materials to be debited back to fuel inventory accounts when reprocessed or held for future use [24]. These provisions reflect the era when reprocessing was expected. Today, with no reprocessing pathway, the expected net salvage value of spent fuel is effectively zero or negative. Re-establishing a credible recycling pathway would give that existing accounting language economic meaning again.

3.3 The French model: Value where use is credible, liabilities are not credible

France offers the most mature example of dual treatment:

- **Legal classification.** French law distinguishes radioactive materials (substances for which a subsequent use is planned or envisaged) from radioactive waste. The State may requalify a material as waste if prospects for use are not sufficiently established, after review by the safety authority [10, 31]. In 2026 the safety authority recommended requalifying substantial parts of some depleted-uranium and thorium-bearing inventories, which shows the classification is tested, not assumed [31].
- **Industrial service.** Orano operates La Hague (licensed for 1,700 tonnes/year; 871 tonnes treated in 2024) and the Melox MOX plant, selling recycling as a commercial service to EDF and foreign customers [9].
- **Conservative valuation, full provisioning.** Independent analysts have documented that EDF historically assigned zero book value to its separated plutonium and reprocessed uranium because of uncertainty over future use, even while recycling them [32]. At the same time, EDF books full provisions for the back end. Its auditors report that at December 31, 2025, EDF's provisions for its French nuclear obligations totaled about EUR 54.3 billion, of which about EUR 31.6 billion related to the back end of the cycle (spent fuel management and radioactive waste) and about EUR 22.8 billion to decommissioning and last cores; under a 2006 French law, EDF must hold dedicated assets (realizable value about EUR 42.5 billion) to secure these obligations [33]. Whether EDF currently assigns any carrying value to recovered plutonium and reprocessed uranium should be confirmed against its latest Universal Registration Document [verify].

The lesson for U.S. policymakers is not that recovered material will automatically appear as a large asset. It is that a functioning recycling pathway changes the character of the obligation. Spent fuel stops being an open-ended storage liability with no exit and becomes feedstock for a defined industrial service with a contractual price, a defined residual waste stream, and a measurable end date. Lenders, rating agencies, and regulators can price a defined service; they cannot easily price an indefinite federal default.

3.4 What would have to change for recovered material to be recognized as an asset

Under U.S. GAAP, an asset requires a present right to economic benefits controlled by the entity. Today, recovered nuclear material fails that test in practice: there is no licensed recycler, no market price, and the Standard Contract contemplates transfer of title to DOE. The following changes would, together, create conditions for recognition:

Table 2. Changes needed to move spent fuel toward the asset column

Domain	Current situation	Change needed	Responsible body
Statute (NWPA)	DOE obligated to accept SNF for disposal; no recycling pathway; NWF limited to disposal	Authorize a voluntary recycling pathway that satisfies DOE acceptance obligations; allow NWF use for residual-waste disposal from recycling	Congress
Standard Contract	Title passes to DOE on acceptance; damages for delay	Amendment option: owner may deliver SNF to licensed recycler, with defined title, credit, and liability release terms	DOE with utilities
GAAP / SEC reporting	No market or contract supports value; costs in AROs and expenses	Contract or observable market price supports recognition (e.g., inventory at lower of cost or net realizable value, or reduced ARO); SEC disclosure guidance on judgments	FASB, SEC staff, auditors
FERC accounting	United States already allows net salvage value; currently zero or negative in practice	Guidance on estimating net salvage value once recycling contracts exist; regulatory asset/liability treatment	FERC
State rate treatment	Storage costs and DOE recoveries flowed through rates; ratepayers paid NWF fees	Rules on sharing recovered value between ratepayers and shareholders; prudence standards for recycling contracts	State PUCs, NARUC
NRC licensing	Prior reprocessing rulemaking discontinued (2021) [34]; one-step Part 70 option proposed (2026)	Final rule with safeguards-by-design, security, and waste-form provisions	NRC
Tax and finance	Recycling facilities' eligibility for federal incentives unclear	Clarify eligibility for loan guarantees and clean-energy incentives [verify current eligibility]	Congress, Treasury, DOE LPO

3.5 Illustrative example economics

ILLUSTRATIVE ONLY: *The figures in this subsection are hypothetical placeholders chosen for arithmetic clarity. They are not estimates of actual costs, prices, or values, and must not be cited as such.*

Consider a hypothetical utility, "Utility X," holding 1,000 MTHM (1,000,000 kgHM) of spent fuel. Let V be the net value per kilogram of heavy metal that a recycler would pay or charge under a contract, after transport and recycling fees and after allocation of residual-waste disposal costs. V can be positive (the recycler pays), zero, or negative (the utility pays a fee, as with disposal today). Let S be the utility's avoided annual on-site storage cost if fuel is removed earlier than it otherwise would be.

Table 3. Hypothetical sensitivity of Utility X's back-end position (illustrative placeholders only)

Scenario	Hypothetical V ($/kgHM)	Balance-sheet effect on 1,000 MTHM	Interpretation
A. Status quo (no pathway)	n/a	No asset; indefinite storage obligation, partly offset by DOE litigation	Liability with no defined end date
B. Recycling at net cost	−100	$100 million defined obligation	Still a liability, but finite and contractible
C. Break-even recycling	0	No asset, no residual obligation beyond contract	Liability eliminated on delivery
D. Recycling at net value	+50	$50 million potential asset (if recognition criteria are met)	Spent fuel moves to the asset column

Two observations follow. First, even Scenario B can improve a utility's financial position relative to Scenario A, because a finite, contractible obligation with a delivery schedule is easier to fund, rate-base, and disclose than an indefinite one; avoided storage costs (S per year, per site) add to that benefit. Second, whether real-world values resemble Scenario C or D depends on factors outside any one utility's control: uranium and enrichment prices, recycling facility capital costs, the value advanced-reactor developers place on recycled feedstock, and the federal policy on residual waste. That is why the accounting change must follow, not precede, the policy and contractual changes in Table 2.

3.6 Addressing the cost critique

Credible analyses have found recycling more expensive than direct disposal under historical conditions [19, 20], and the NEA found back-end costs to be a relatively small share of total generation costs, with results sensitive to discount rates, fast-reactor costs, and uranium prices [29]. This paper's argument does not depend on recycling being cheaper per kilogram in every scenario. It rests on three points: (1) the realistic U.S. alternative is not low-cost direct disposal but indefinite storage plus growing Judgment Fund payments; (2) the advanced-reactor fleet now being planned creates a demand for recycled feedstock that did not exist when earlier studies were written; and (3) new technology programs such as ARPA-E CURIE explicitly target lower-cost, more proliferation-resistant approaches [8]. Proponents should commit to transparent, independently reviewed cost estimates at each phase.

4. Reducing Long-Term Environmental Risk

Scope note: *As elsewhere in this paper, this section is conceptual and policy-level. Time scales and reduction factors are those published by public bodies and companies; they depend on assumptions and are shown only to indicate orders of magnitude.*

The environmental case for recycling is not that it eliminates nuclear waste. It does not. The case is that recycling changes the form, volume, and duration of the hazard that society must manage. It also replaces open-ended storage of intact spent fuel at dozens of sites with a smaller amount of conditioned waste destined for geologic disposal.

4.1 A smaller, more stable, more tightly controlled waste form

In the French system, Orano reports that about 96 percent of used fuel (uranium and plutonium) is recyclable and that the remaining roughly 4 percent, mostly fission products, is immobilized in glass and sealed in stainless-steel canisters stored at La Hague pending geologic disposal [35]. Orano states that this reduces the volume of the most radioactive waste by a factor of about 5 and its long-term radiotoxicity by a factor of about 10 [36]. Orano has also acknowledged in France's public debate that the volume factor assumes spent MOX fuel will itself be recycled later, and that it falls to "more than 3" if spent MOX is counted as waste [37]. That honest range is still a material reduction. Vitrified glass is a stable, engineered waste form designed for disposal. Concentrating on a small number of controlled facilities is a different risk profile from intact assemblies dispersed across many sites.

4.2 Shorter-lived radiotoxicity once long-lived actinides are recycled

Most of spent fuel's long-term radiotoxicity comes from plutonium and the other long-lived actinides, not from the fission products that dominate the first few centuries. A European Commission Joint Research Centre report, summarizing OECD-NEA work, describes the time for waste radiotoxicity to fall to the level of the natural uranium ore originally mined [38]:

Table 4. Approximate time for waste radiotoxicity to fall to the level of the original uranium ore (published conceptual estimates)

Fuel-cycle option (conceptual)	Approximate time to reach ore level	Source
Open (once-through) cycle: spent fuel disposed directly	More than 200,000 years (IAEA analysis cites about 130,000 years under different assumptions)	[38, 39]
Partially closed: single recycle of plutonium	About 100,000 years	[38]
Fully closed: multiple recycle of plutonium	About 30,000 years	[38]
Fully closed with recycle of minor actinides as well (partitioning and transmutation)	Theoretically about 400 years, or under 1,000 years; some wastes still require long isolation	[38, 40]

The pattern is the point: from hundreds of thousands of years toward hundreds to a few thousand years, but only as the system matures from single recycling toward fast-reactor multi-recycle of actinides. Three caveats keep the claim credible. First, radiotoxicity inventory is a measure of hazard potential, not of actual risk; a well-sited repository is designed to keep doses low in either case. Second, some long-lived fission products are not reduced by actinide recycling, and they remain a repository design consideration [38]. Third, the largest gains require reactors that consume a broad range of actinides, which is why this paper links recycling to the advanced-reactor fleet.

4.3 The risks of the status quo: storage without an end date

Today, spent fuel sits in pools and dry casks at about 75 sites [21]. Dry storage is safe and well regulated today. The NRC licenses storage systems for terms of up to 40 years at a time, with aging-management requirements at renewal [41]. The issue is the time horizon, not present-day safety. The NRC's Continued Storage environmental analysis considered an indefinite-storage scenario. That scenario assumes institutional controls continue and that casks, storage pads, and transfer facilities are replaced roughly once every 100 years [42]. Over the tens of thousands of years that unrecycled spent fuel remains hazardous, that assumption implies on the order of hundreds of repackaging cycles, each with handling, worker dose, and cost.

- **Aging management.** Canisters, concrete overpacks, and fuel cladding must be monitored and eventually repackaged; every cycle depends on future funding and expertise.
- **Institutional control.** Indefinite storage assumes stable governance, regulation, and funding over periods far longer than any institution has existed.
- **Climate and sitting.** Many reactor sites were chosen for access to cooling water, including coastal, riverine, and lakeshore locations; long-term exposure to flooding, storm surge, and shoreline change is a legitimate planning concern [verify site-specific assessments].
- **Community burden.** Host communities, especially at shutdown plants, carry land-use restrictions, security, and stigma for fuel they were promised would be removed [17].

4.4 Counterpoints and how modern regulation manages them

Reviewers will rightly ask what recycling adds to environmental risk. The honest answers are these:

- **Secondary waste streams.** Recycling facilities generate intermediate- and low-level waste and, eventually, decommissioning waste. These must be counted in any comparison, and the BRC cautioned that no foreseeable technology eliminates the need for geologic disposal [43].
- **Plant discharges.** Recycling plants release some radio nuclides to air and water under permit. OSPAR data show that La Hague accounted for the large majority of reported tritium discharges from nuclear facilities in the OSPAR region in 2024 [44]. OSPAR's assessments have generally not found significant radiological impact [verify]. A U.S. facility would be subject to EPA's environmental radiation standards for the uranium fuel cycle (40 CFR Part 190), which date from 1977 and include release limits relevant to recycling; EPA should review them for modern facilities [45] [verify scope of applicability].
- **Transport.** Recycling adds shipments of fuel, recovered materials, and waste. These are governed by NRC and DOT packaging rules and, for sensitive materials, by NRC security requirements. Colocation of facilities is the most effective way to minimize transport.
- **Accident and security consequences.** Concentrated inventories at a recycling facility require robust safety and security design, which NRC licensing and safeguards-by-design are intended to provide [15, 25].

The policy recommendation is not to assume the environmental benefit but to demonstrate it. A programmatic environmental impact statement for a U.S. recycling program should compare, on a life-cycle basis, the recycling pathway against indefinite storage plus direct disposal, including secondary wastes, discharges, transport, and long-term institutional risk.

5. Strengthening Nonproliferation Through Recycling and MOX

Scope note: *This section addresses policy arguments only. It contains no information on processes, material properties, facility design, or safeguards and security measures beyond the existence of public regulatory requirements.*

5.1 The case: Burn the plutonium rather than bury it

Every year of once-through operation adds plutonium to the U.S. spent fuel inventory. In the near term, that plutonium is protected by the intense radiation of the surrounding fission products. Over centuries, however, that radiation barrier declines while the plutonium remains. Analysts have long discussed whether repositories could, in the distant future, become "plutonium mines" [46]. Recycling plutonium into mixed-oxide (MOX) fuel or advanced-reactor fuel, and fissioning it in reactors, converts a growing long-term liability into electricity and can reduce the plutonium inventory over time, particularly with multi-recycle in fast reactors [29].

A second argument is economics. Material that has value is accounted for, tracked, and protected as an asset. Facilities handling significant quantities of plutonium in the United States are subject to the NRC's most stringent physical protection and material control and accounting requirements for strategic special nuclear material, commonly referred to as Category I [47, 48]. They can also be placed under IAEA safeguards through the U.S. voluntary offer agreement. Recycling places the material inside an active, inspected, and continuously accounted system. Direct disposal relies on burial and the passage of time.

5.2 The standard critique, stated fairly

The counterargument is serious and has shaped U.S. policy since 1977. The Union of Concerned Scientists argues that reprocessing separates weapon-usable plutonium from the protective radiation of spent fuel, increasing risks of theft and diversion. UCS also argues that MOX does not provide meaningful protection because it can itself be processed [49]. Harvard's Belfer Center analysts concluded that reprocessing was more costly than direct disposal and raised security concerns [20]. International experience supports the concern that separation can outpace use. The United Kingdom accumulated about 140 tonnes of civil separated plutonium and in 2025 decided to immobilize it for disposal rather than use it [50].

The U.S. MOX project is a cautionary tale of a different kind. Under the U.S.-Russia Plutonium Management and Disposition Agreement, each side committed to dispose of at least 34 metric tons of surplus weapons plutonium [51]. The U.S. chose MOX, but the Savannah River facility suffered major cost growth and was terminated in 2018 [52, 53]. Russia suspended the agreement in 2016, citing in part the U.S. shift toward a dilute-and-dispose approach at the Waste Isolation Pilot Plant [51, 54]. The lesson is that a plutonium-use program must be affordable and executable, or it will be abandoned midstream, leaving the material unresolved.

5.3 The policy answer

The critique is an argument against separated plutonium stockpiles and poorly executed programs. It is not an argument against every form of recycling. A U.S. program can be designed to answer it directly:

- **No stockpiling.** Adopt as binding policy that no civil separated plutonium may accumulate beyond working inventories. Any recovered plutonium must be committed to fuel on a defined schedule.
- **Balanced separation and use.** Size recovery to match fuel demand from operating reactors so that inventories do not build up, which is the failure mode seen in the UK.
- **Co-processing and group recovery.** Prefer approaches in which plutonium is never produced as a pure product, consistent with ARPA-E CURIE's emphasis on reducing material attractiveness [8].
- **Colocation.** Co-locate recycling, fuel fabrication, and, where practical, reactors on secured sites to minimize transport, as DOE's federal-site approach contemplates [12, 13].
- **Safeguards-by-design and Category I security.** Build IAEA-guided safeguards and NRC Category I protection into facility design from the start [25, 47, 55].
- **Transparency and leadership.** Continue and expand public reporting of civil plutonium holdings under the IAEA's Guidelines for the Management of Plutonium (INFCIRC/549) [56]. Use U.S. leadership, through 123 agreements and supplier coordination, to make no-stockpile, safeguarded, co-located recycling the international norm. Pair it with fuel-service offers so that partners have no need for their own sensitive facilities.

Framed this way, recycling becomes a nonproliferation tool. It reduces the long-term plutonium inventory, holds material under active accounting and inspection, and gives the United States standing to shape global norms. Today it cedes that role to countries with large existing programs.

6. Reducing Long-Term Federal Expenditures

6.1 What the current model costs the government

Under the NWPA and the Standard Contract, DOE must take title to commercial spent fuel and dispose of it. Because it has not, taxpayers pay twice: through damages for the delay and, eventually, through the disposal program itself.

- **Judgment Fund damages.** About $12.2 billion has been paid through September 30, 2025, and the remaining liability is estimated at $38.6 billion to $44.3 billion, for a total of $50.8 billion to $56.5 billion [6]. The Nuclear Waste Strategy Coalition testified that liability grows by about $2 million per day [17]. CBO has emphasized that these costs are borne by taxpayers through the Judgment Fund and rise with each year of delay [57].
- **Repository cost.** DOE's last published Total System Life Cycle Cost estimate (2008) put the disposal program at $96.18 billion in constant 2007 dollars from 1983 through closure in 2133. That total comprised about $64.7 billion for the repository, $20.3 billion for transportation, and $11.2 billion for other program costs, for 122,100 MTHM of spent fuel

and high-level waste [58]. That estimate is now dated [verify: no updated TSLCC published since 2008].

- **Repository capacity.** The NWPA limits the first repository to 70,000 MTHM until a second repository is operating, and DOE has stated that there is no cost estimate, design, or authorization for a second repository [59]. The current commercial inventory of 95,117 MTHM [3] already exceeds that statutory limit.
- **Interim storage.** Federal consolidated interim storage, which DOE is pursuing through consent-based siting, would add facility, transport, and operations costs; no current official life-cycle estimate is cited here [verify].

6.2 Mechanisms by which recycling could reduce federal costs

1. Lower Judgment Fund accrual. A recycling pathway that lets utilities deliver fuel to licensed recyclers, with negotiated contract releases, would begin satisfying federal obligations sooner than a repository can, slowing or stopping accrual of damages.
2. Smaller or fewer repositories. Removing heat-generating actinides can, in idealized analyses, substantially increase how densely waste can be replaced. A NEA review cited roughly a six-fold increase in repository loading from removing plutonium and americium [40]. That could reduce the size of the first repository or the need for a second.
3. Private capital. DOE's April 2026 recycling solicitations state that selected offerors will be fully responsible for project costs [13], shifting capital risk from taxpayers to investors.
4. Revenue from recovered fuel and federal assets. Recovered material sold as fuel, leases of federal land, and fees for recycling services can offset program costs, with any value-sharing defined by statute and contract.
5. Reduced indefinite storage. Fewer decades of at-reactor storage mean fewer repackaging cycles and lower long-run federal reimbursement exposure.

These benefits are not automatic. Recycling facilities are expensive, and if they require large federal subsidies, net federal savings could shrink or disappear. The BRC concluded in 2012 that it was premature to commit to a closed cycle and that geologic disposal is needed in any case [22, 43]. GAO has stressed that congressional action on the NWPA and the Nuclear Waste Fund is the precondition for progress [21]. The right test is a transparent, CBO-scored comparison of the recycling pathway against the realistic baseline of continued damage, interim storage, and direct disposal.

ILLUSTRATIVE ONLY: *Table 5 applies hypothetical shares and reductions to verified baseline figures purely to show orders of magnitude. The percentages are assumptions, not estimates, and the results are not forecasts.*

Table 5. Illustrative federal cost levers (verified baselines; hypothetical shares)

Lever	Verified baseline	Hypothetical assumption	Illustrative result	Caveat
Judgment Fund accrual	~$2 million per day [17] (about $0.73 billion per year, derived)	Recycling pathway halts accrual for 25%-100% of the obligation	About $0.18-$0.73 billion avoided per year	Accrual is not linear; depends on settlements [verify]
Remaining liability	$38.6-$44.3 billion [6]	10%-30% retired through contract releases	About $3.9-$13.3 billion	Requires NWPA and contract changes
Repository (2007$)	$64.7 billion repository element of $96.18 billion TSLCC [58]	Heat-driven costs reduced 10%-30%	About $6.5-$19.4 billion (2007$)	Dated estimate; recycling wastes still need disposal
Second repository	No design or estimate exists [59]	Recycling avoids or defers need	Not quantifiable	Major potential saving; unverifiable today
Capital risk	DOE RFAs: offerors bear project costs [13]	Private financing of first facilities	Qualitative	Depends on final contract terms

7. The Historical Record

The United States pioneered reprocessing, for defense purposes, and briefly attempted a commercial industry before abandoning it on policy grounds.

- **Early commercial efforts.** Nuclear Fuel Services operated the West Valley, New York plant from 1966 to 1972, the only U.S. commercial reprocessing plant ever to operate; it processed about 640 metric tons of fuel [60]. General Electric's Morris, Illinois, never operated commercially; the facility's pool is still used for spent fuel storage, under an NRC license renewed in 2022 through May 31, 2042 [61, 62]. Allied-General Nuclear Services' Barnwell, South Carolina plant was built but never operated commercially [61, 63].
- **Ford's deferral (1976).** In October 1976, President Ford announced that the United States should no longer regard reprocessing as a necessary and inevitable step, deferring commercial reprocessing pending a nonproliferation review [63].
- **Carter's halt (April 1977).** On April 7, 1977, President Carter announced an indefinite deferral of commercial reprocessing and plutonium recycling on nonproliferation grounds, intended to set an international example [63].
- **Clinch River Breeder Reactor.** President Carter sought to cancel the Clinch River Breeder Reactor project, but Congress continued to fund it. The project ended when the Senate voted

in October 1983, under President Reagan, to deny further funding. Only site preparation and excavation were completed; no construction permit was issued [64].

- **Reagan lifts the ban (1981).** In October 1981, President Reagan lifted the indefinite ban on commercial reprocessing, but no company restarted commercial reprocessing, owing to cost, policy uncertainty, and lack of federal support [63].
- **Integral Fast Reactor (1994).** Argonne National Laboratory's Integral Fast Reactor program, which paired a fast reactor with on-site fuel recycling, was terminated by Congress in 1994 at the urging of the Clinton administration [65].
- **GNEP (2006-2009).** The Global Nuclear Energy Partnership, launched in 2006, sought to develop proliferation-resistant recycling with international partners. GAO questioned DOE's accelerated approach in 2008, and DOE cancelled GNEP's programmatic environmental impact statement in June 2009, ending the domestic commercial recycling focus [66, 67].
- **MOX Fuel Fabrication Facility (2018).** Construction of the MOX plant at the Savannah River Site, intended to convert surplus weapons plutonium into reactor fuel under a U.S.-Russia agreement, began in 2007. Citing cost growth, NNSA terminated the contract on October 10, 2018, and the NRC terminated the construction authorization in February 2019 [52, 53].
- **Renewed interest (2022-2026).** ARPA-E launched CURIE in 2022, awarding about $38 million to 12 initial projects on recycling, materials accountancy, and facility design [8, 68]. Executive Orders 14299 and 14302 (May 2025) directed DOE to recommend a national fuel-cycle policy and to authorize private recycling on federal sites [11, 12]. Oklo announced a recycling facility in Tennessee in September 2025 [16]. DOE issued recycling RFAs in April 2026 [13], and the NRC proposed a one-step licensing option in June 2026 [15].

Table 6. Timeline of U.S. reprocessing and recycling policy

Year	Event	Outcome
1966-1972	West Valley, NY plant operates (Nuclear Fuel Services)	Only U.S. commercial plant to operate; ~640 t processed
Early 1970s	Morris, IL (GE) and Barnwell, SC (AGNS) built	Neither operated commercially
1976	President Ford defers commercial reprocessing	Policy paused on nonproliferation grounds
Apr. 1977	President Carter halts commercial reprocessing	Indefinite deferral; Barnwell abandoned
1977-1983	Carter seeks to cancel Clinch River Breeder Reactor; Congress keeps funding	Senate ends funding Oct. 1983; only site preparation done
1981	President Reagan lifts the ban	No industry restart
1982 / 1987	NWPA enacted; amendments designated Yucca Mountain	Federal disposal obligation; 1998 acceptance date
1994	Integral Fast Reactor program terminated	End of U.S. integrated fast-reactor recycling R&D
1998	DOE misses Standard Contract acceptance date	Utility litigation begins

Year	Event	Outcome
2006-2009	Global Nuclear Energy Partnership	PEIS cancelled June 2009
2010	Yucca Mountain defunded; DOE seeks to withdraw application	Impasse continues
2014	NWF fee set to zero (May 16)	Fund balance stranded
2018	MOX Fuel Fabrication Facility terminated	No U.S. plutonium-fuel capability
2022	ARPA-E CURIE awards (~$38 million)	Recycling R&D restarts
2025	Executive Orders 14299 and 14302; Oklo Tennessee announcement	Federal policy turns toward recycling
2026	DOE recycling RFAs (Apr.); NRC proposed rule (June); congressional hearings (Sept.)	Framework under construction

8. International Comparison

Several nations recycle, have recycled, or are building fast-reactor systems. Their experience offers both models and caution.

Table 7. International comparison (summary)

Country	Status	Lesson for the United States
France	Orano La Hague: licensed 1,700 t/yr, 871 t treated in 2024; Melox MOX plant; MOX ~10% of nuclear generation [9]	Recycling can be a stable industrial service; legal "materials vs. waste" classification is tested periodically [31]
Japan	Rokkasho reprocessing plant completion repeatedly delayed (27th postponement in 2024, target FY2026); further delay reported likely [69, 70]	Schedule and cost discipline matter; first-of-a-kind complexity is real
United Kingdom	THORP oxide reprocessing ended 2018; Magnox reprocessing ended 2022 [71]; 2025 decision to immobilise ~140 t of civil plutonium for disposal [50]	Separation without a reuse plan creates a stockpile liability; avoid separating materials without a use
Russia	Operates reprocessing and fast reactors, and is developing integrated fast-reactor fuel-cycle projects [72]	Integration of recycling with fast reactors is achievable; competitive pressure on U.S. leadership
India	Closed cycle central to its three-stage program; 500 MWe Prototype Fast Breeder Reactor reached first criticality Apr. 6, 2026 [73]	Strategic, long-horizon commitment to fuel-cycle self-reliance

The comparison supports a clear design rule for the United States: recycle only with a defined use for recovered material, avoid accumulating separated plutonium, and integrate recycling with the reactors that will consume its products. The United Kingdom's experience shows the financial and security cost of separating material without a reuse plan; France shows the value of a coherent, long-term industrial program; Japan shows the risks of first-of-a-kind schedules.

9. Policy and Political Changes Needed

No single action will move spent fuel to the asset column. The changes below are designed to work as a package across five domains.

Table 8. Policy-change matrix

Domain	Specific change	Lead	Priority
Legislative	Amend NWPA to authorize a voluntary recycling pathway that satisfies DOE acceptance obligations; define disposal of recycling residues	Congress (Energy & Commerce; EPW; ENR)	High
Legislative	Reform NWF budget treatment so fees and balances can fund integrated back-end programs [21, 22]	Congress, OMB	High
Legislative	Consider a dedicated waste-management organization outside DOE, as the BRC and BPC recommended [22, 23]	Congress	Medium
Regulatory	Finalize NRC licensing option for recycling facilities with safeguards-by-design and security requirements [15]	NRC	High
Regulatory	FERC guidance on net salvage value and regulatory asset/liability treatment [24]	FERC	Medium
Regulatory	State PUC frameworks for sharing recovered value and approving recycling contracts	State PUCs, NARUC	Medium
Executive	Complete EO 14302 policy and implement DOE recycling RFAs with transparent milestones [11, 13]	DOE, NNSA	High
Executive	Standard Contract amendment template for recycling deliveries	DOE, DOJ	High
International	Commit to IAEA safeguards eligibility for commercial recycling; coordinate with France, Japan, UK, and allies through 123 agreements	State, DOE, NNSA	High
Financial	Clarify loan-guarantee and incentive eligibility; offtake or fuel-bank mechanisms for recycled feedstock	DOE LPO, Treasury	Medium

Domain	Specific change	Lead	Priority
Financial	Accounting guidance (FASB/SEC staff, auditors) on recognition once contracts exist	FASB, SEC, industry	Medium
Environmental	Programmatic EIS comparing recycling, continued storage, and direct disposal (Section 4)	DOE, NRC, EPA	High
Federal fiscal	Recycling-pathway credits to reduce Judgment Fund exposure; repository sizing on heat and performance (Section 6)	Congress, DOE, DOJ	High
Citation-level reforms	Specific statutes, rules, and policies in Tables 9-11 (Section 10)	Congress, NRC, DOE, EPA, White House, States	High
Export controls	Part 810 and Part 110 list reviews, 123 renewals, and cooperative authorizations for partners (Section 11)	DOE, NRC, State, Commerce	High

10. Statutory, Regulatory, and Policy Changes Required

Section 9 summarized the needed changes by domain. This section is a citation-level inventory for legislative counsel, regulators, and reviewers. It identifies each statute, regulation, executive policy, and international instrument that bears on restarting a U.S. closed fuel cycle. For each, it states what it does now, why it is or is not a barrier, the recommended action, the responsible body, and the nonproliferation protections that would be retained. Many items need no change at all. Where that is the case the tables say so plainly, because a credible reform agenda should be precise about what it does not seek to weaken.

How to read Tables 9-11: *Actions: Revise (amend existing text); Eliminate (repeal or rescind); New (new statute, rule, or policy); Retain (no change; comply). Characterizations of legal text are policy-level summaries for discussion, not legal opinions; counsel should confirm each before drafting.*

10.1 Statutes, executive policies, and non-NRC regulations

Table 9. Statutes, executive policies, EPA/DOE regulations, and state laws

Citation	What it does now	Why it matters	Action	Lead	Safeguards retained
NWPA of 1982, as amended 1987, 42 U.S.C. 10101 et seq. [74]	Federal responsibility for disposal; Yucca as sole candidate site; defines HLW to include waste from reprocessing	No recycling pathway; source-based HLW definition makes all recycling residues HLW regardless of hazard [verify interpretation]	Revise: add recycling pathway; consider risk-informed waste definitions	Congress	Repository isolation standards unchanged
NWPA Sec. 114(d), 42 U.S.C. 10134(d) [59, 74]	70,000 MTHM cap on first repository until a second operates	Current 95,117 MTHM commercial inventory already exceeds cap [3]	Revise: base capacity on heat/performance, not mass	Congress	NRC/EPA performance standards
NWPA Sec. 302, 42 U.S.C. 10222; NWF fee suspension (2014) [26, 27]	Fee and Nuclear Waste Fund; fee set to zero since May 16, 2014	Fund stranded by budget rules; no fee for new program	Revise: NWF access reform; fee tied to delivered service incl. recycling	Congress, OMB	Ratepayer funds dedicated to back end
Standard Contract, 10 CFR Part 961 [5]	DOE takes title to SNF for disposal; 1998 start date	No option to deliver to a recycler or retain title to recovered value	Revise: amendment option for recycling deliveries, credits, releases	DOE, utilities	DOE retains backstop obligation

Citation	What it does now	Why it matters	Action	Lead	Safeguards retained
DOE 10 CFR Part 960 [75]	Repository site-screening guidelines under NWPA	Written for direct disposal of spent fuel	Revise: update for consent-based siting and recycling waste forms [verify current use]	DOE	Sitting protective criteria
AEA Sec. 11v and Sec. 103/104, 42 U.S.C. 2014, 2133-2134 [76]	Reprocessing plants are "production facilities" requiring NRC licensing	Two-step licensing historically under Part 50	Retain statute; implement via NRC rule (see Table 10)	NRC	Full NRC safety and security review
AEA Sec. 123, 42 U.S.C. 2153 [76]	Agreements for cooperation with nonproliferation criteria; congressional review	Partner agreements govern foreign-origin material and fuel services	Retain; negotiate or update agreements (Section 11)	State, DOE, Congress	All Sec. 123 criteria retained
AEA Sec. 131, 42 U.S.C. 2160 [76, 77]	Subsequent arrangements, incl. U.S. consent to reprocessing of U.S.-obligated material	Needed for any international recycling services	Retain; use programmatic consent with qualifying partners	DOE, State	Case-by-case or programmatic consent
AEA Sec. 170 (Price-Anderson), 42 U.S.C. 2210; 10 CFR Part 140 [76, 78]	Liability and indemnity framework for licensed nuclear activities	Reprocessing plants are production facilities, so coverage is mandatory, but Part 140 specifies no financial-protection amount or fee for them today [78, 79]	Revise: set financial-protection levels and fees for recycling facilities (none specified today), conforming to any Part 70 licensing rule [15]	Congress, NRC	Public compensation assured

Citation	What it does now	Why it matters	Action	Lead	Safeguards retained
Nuclear Non-Proliferation Act of 1978, 22 U.S.C. 3201 et seq. [80]	Export criteria, safeguards conditions, consent rights; policy to discourage spread of sensitive facilities	Does not prohibit domestic recycling; shapes international cooperation	Retain	Congress	All export criteria retained
NEPA, 42 U.S.C. 4321 et seq.; CEQ regulations removed (eff. Apr. 11, 2025); DOE procedures revised (July 2025) [81, 82, 83]	Environmental review of major federal actions	Procedures in flux; recycling needs a durable record	New: programmatic EIS for a national recycling program	DOE, NRC	Full environmental disclosure
NEIMA (Pub. L. 115-439) and ADVANCE Act (Pub. L. 118-67) [84]	Modernize NRC licensing; fees; advanced reactors	Focused on reactors; little on fuel-cycle facilities	Revise: extend modernization mandates to recycling facilities [verify NEIMA citation]	Congress	NRC independence
EPA 40 CFR Part 190 [45]	Uranium fuel cycle environmental radiation standards, incl. release limits relevant to recycling	Dates from 1977; compliance basis for modern facilities untested [verify]	Revise: EPA review and update	EPA	Public dose limits retained
EPA 40 CFR Part 191 [85]	Generic disposal standards for SNF, HLW, TRU	Would govern a non-Yucca repository receiving recycling waste	Retain; confirm applicability to vitrified recycling waste	EPA	Long-term release limits
EPA 40 CFR Part 197 [86]	Yucca Mountain-specific standards	Site-specific; not a barrier to recycling	Retain (no change needed)	EPA	Unchanged

Citation	What it does now	Why it matters	Action	Lead	Safeguards retained
Carter policy statement, Apr. 7, 1977 [63]	Indefinite deferral of commercial reprocessing	Superseded in 1981; still cited as precedent	No action (superseded); note in new policy	White House	--
Reagan policy statement, Oct. 1981 [63]	Lifted indefinite ban on commercial reprocessing	Supportive; no implementing program	Retain; build on	White House	--
PDD-13 policy (Sept. 27, 1993) [87]	U.S. "does not encourage the civil use of plutonium" and does not itself reprocess for power	Most direct executive-policy barrier; no public rescission identified [verify current status]	Eliminate/replace new presidential fuel-cycle policy with no-stockpile and safeguards principles	White House, NSC	Plutonium minimization goals retained
EO 14299, EO 14300, EO 14302 (May 23, 2025) [11, 12, 88]	Direct DOE recycling policy, federal-site recycling, NRC reform	Supportive; executive orders can be reversed	Retain; codify key elements in statute	White House, Congress	Safeguards and security requirements
MOX Fuel Fabrication Facility termination (Oct. 10, 2018) [52, 53]	Ended weapons-plutonium MOX path; dilute-and-dispose adopted [54]	Lesson on cost discipline; separate from commercial recycling	Retain decision; keep defense plutonium disposition separate	DOE/ NNSA	Surplus plutonium disposition continues
NRC reprocessing rulemaking discontinued, SECY-21-0026 [34]	Ended reprocessing-specific rulemaking	Left no tailored framework until 2026 proposal	Superseded by 2026 proposed rule (Table 10)	NRC	--
State moratoria, e.g., Cal. Pub. Res. Code Sec. 25524.2 [89]	Bars new nuclear plants until a federally approved HLW disposal means exists	Ties new build to the back-end impasse	Revise (state level): recognize licensed recycling plus disposal pathway; other states' laws [verify]	State legislatures	State siting authority retained

10.2 NRC regulations (10 CFR)

The NRC framework is largely adequate. Most parts require no change, and the key need is a finalized licensing pathway for recycling facilities. The NRC's proposed June 2026 rule would allow such facilities to use a one-step Part 70 license [15], building on Part 53, which took effect April 29, 2026 for reactors [90].

Table 10. NRC regulations: change needed or not

10 CFR Part	Current function	Change needed?	What would change	Reference
Part 20	Radiation protection standards	No change	Applies as is to recycling facilities	[91]
Part 50	Licensing of production and utilization facilities; reprocessing plants historically licensed here	Revise (conforming)	Retain as option; conform to Part 70 one-step alternative	[15, 92]
Part 51	NEPA implementation; Continued Storage rule (Sec. 51.23)	Revise (targeted)	Generic or programmatic EIS for recycling; review fuel-cycle impact tables [verify]	[42, 93]
Part 52	Combined licenses and design certification for power reactors	No change	Reactors using recycled fuel licensed normally	[94]
Part 53	Technology-inclusive reactor licensing (eff. Apr. 29, 2026)	No change anticipated	Confirm fuel-qualification pathways for recycled fuels [verify]	[90]
Part 60	Generic HLW geologic disposal	Revise	Modernize for non-Yucca repository and recycling waste forms	[95]
Part 61	Land disposal of LLW; waste classification	Clarify	Classification of recycling secondary wastes and GTCC [verify rulemaking status]	[96]
Part 63	Yucca Mountain-specific disposal	No change	Not a barrier to recycling	[97]
Part 70	SNM licensing, incl. integrated safety analysis	Revise (finalize 2026 proposal)	One-step license for recycling facilities with safeguards-by-design	[15, 98]
Part 71	Packaging and transport	No change	Compliance; existing package certification	[99]

10 CFR Part	Current function	Change needed?	What would change	Reference
Part 72	Independent storage of SNF, HLW, reactor-related GTCC	No change (minor clarifications possible)	Storage of vitrified waste pending disposal [verify]	[100]
Part 73	Physical protection, incl. Category I (Sec. 73.46)	No weakening; conforming updates	Apply Category I to recycling facilities as applicable	[47]
Part 74	MC&A, incl. Category I (Subpart E)	No weakening; conforming updates	Accounting for grouped or co-recovered materials [verify]	[48]
Part 75	Implementation of U.S.-IAEA safeguards	No change	Facility placed on eligible list; reporting	[101, 102]
Part 110	Export/import licensing	No change for domestic program	Exports for partners follow existing criteria (Section 11)	[103]
Part 140	Price-Anderson financial protection	Revise	Set financial protection amounts and indemnity fees for recycling facilities; none are specified today (Part 140 does set them for plutonium processing and fuel fabrication plants)	[78, 79]
SECY-21-0026; 2026 proposed rule	Prior reprocessing rulemaking discontinued; new Part 70 option proposed	Finalize	Final rule; comment period closed Aug. 2026 [verify final date]	[15, 34]

10.3 International treaties and agreements

Table 11. International instruments: revision or compliance

Instrument	What it requires	Revision needed?	U.S. body
NPT (1970) [104]	Arts. III-IV: safeguards (for non-nuclear-weapon states) and right to peaceful use; U.S. is a nuclear-weapon state	No; compliance	State

Instrument	What it requires	Revision needed?	U.S. body
U.S.-IAEA Voluntary Offer Agreement, INFCIRC/288 (in force Dec. 1980) [102]	IAEA may apply safeguards at eligible U.S. civil facilities	No; add recycling facilities to eligible list	State, NRC, DOE
Additional Protocol, INFCIRC/288/Add.1 (in force Jan. 6, 2009) [105]	Expanded declarations and access	No; compliance and reporting	State, NRC, DOE, Commerce
CPPNM and 2005 Amendment (in force May 8, 2016; U.S. ratified July 31, 2015) [106, 107]	Physical protection of nuclear material and facilities	No; implemented via 10 CFR 73	NRC, DOE
Joint Convention, INFCIRC/546 (U.S. in force July 14, 2003) [107, 108]	Safety of spent fuel and waste management; spent fuel at reprocessing facilities excluded unless declared	No; decide whether to declare reprocessing in national report	DOE, NRC, State
U.S.-Euratom 123 (in force Apr. 12, 1996; rolling 5-year terms) [109, 110]	Reciprocal long-term consent for specified reprocessing; annex lists U.S. facilities	Update U.S. annex to list new facilities [verify procedure]	State, DOE
U.S.-Japan 123 (1988; in force until terminated) [77, 110]	Programmatic consent to reprocessing in Japan	No; coordinate plutonium balance	State, DOE
U.S.-ROK 123 (2015-2040) [111]	No advance consent for domestic ROK reprocessing; consultation mechanism	No; policy coherence needed	State, DOE
Other 123 agreements; supplier-state consent [110]	Foreign-obligated uranium in U.S. fuel may carry partner consent rights	Review; obtain consents where required [verify per agreement]	State, DOE
PMDA (2000; amended 2010; suspended by Russia 2016) [51]	34 t each of surplus weapon's plutonium	No; keep separate from commercial recycling	State, NNSA
NSG Guidelines, INFCIRC/254 Part 1 [112]	Restraint and criteria for transfers of reprocessing facilities and technology	No; U.S. should champion	State, DOE, NRC, Commerce
Zangger Committee trigger list [113]	Common understandings on export conditions	No	State

Instrument	What it requires	Revision needed?	U.S. body
CSC (in force Apr. 15, 2015; U.S. ratification deposited May 21, 2008) [107, 114, 115]	Supplementary compensation for nuclear damage	No; CSC Annex definition of nuclear installation covers reprocessing and storage installations [115]	State, DOE
INFCIRC/549 plutonium guidelines [56]	Annual public reporting of civil plutonium	No; continue and expand transparency	DOE, State

11. Building a Global Closed Fuel Cycle Cooperative: International Policy, Agreements, and Export Controls

A U.S. closed fuel cycle will source uranium abroad, partner with allied fuel-cycle states, and eventually offer fuel services. It can only do business lawfully within the U.S. nuclear export control regime. This section sets out that regime, applies it country by country, and proposes a cooperative framework. The framework is built on export-control compliance and nonproliferation commitments, not around them.

11.1 The U.S. nuclear export control regime

- 10 CFR Part 810 (DOE): assistance to foreign atomic energy activities. Assistance to destinations listed in Appendix A is generally authorized for most activities [1]. Other destinations require specific authorization by the Secretary. Assistance involving reprocessing or fabrication of plutonium-bearing fuel requires specific authorization for any country, including generally authorized destinations [116]. AEA Sec. 57b is the statutory basis [verify section].
- 10 CFR Part 110 (NRC): export and import of nuclear equipment and material, through general licenses (Secs. 110.20-110.27) and specific licenses with Executive Branch review (Secs. 110.40-110.45). It lists embargoed destinations (Cuba, Iran, Iraq, North Korea, Syria, Sudan) and restricted destinations (Sec. 110.29) [2, 103].
- **AEA Secs.** 109, 123, and 126-129: component exports, agreements for cooperation, export licensing procedures and criteria, and conduct resulting in termination of exports [76]; NNPA 1978 export criteria [80].
- EAR (15 CFR 730-774, Commerce/BIS): nuclear-nonproliferation (NP) controls dual-use items (Sec. 742.3), nuclear end-use and end-user controls (Sec. 744.2), and the Entity List (Supplement No. 4 to Part 744) [117, 118]. The interagency Subgroup on Nuclear Export Coordination reviews nuclear-related dual-use cases [verify citation].
- ITAR (22 CFR 120-130): relevant only where defense articles, such as naval or national-security reactor programs, are involved [119].
- OFAC sanctions (31 CFR Chapter V), including the Democratic Republic of the Congo Sanctions Regulations (Part 547), and Russia-related measures. Separately, the Prohibiting

Russian Uranium Imports Act bans Russian low-enriched uranium imports, with waivers ending by January 1, 2028 [120, 121].

- Multilateral regimes: NSG Guidelines and the Zangger Committee trigger list [112, 113].

11.2 Country-by-country legal basis

Table 12 summarizes the current legal basis for cooperation with key uranium suppliers and fuel-cycle partners. Part 810 status is taken from Appendix A as amended through June 2026 [116]. Part 110 status is from Secs. 110.28-110.30 [103]. 123 status is from DOE/NNSA's list as of July 2025 [110]. Additional Protocol dates are from the IAEA [122], and Pelindaba ratifications are from the African Union [123].

Table 12. Legal basis for cooperation with supplier and partner countries

Country	Role	123 agreement	Part 810	Part 110	NPT / IAEA AP / Pelindaba	Gaps and needed steps
Namibia	Third-largest producer (12% of world output, 2024) [124]	None	Specific authorization required (not in App. A)	Not embargoed/restricted; not in Sec. 110.30 NSG list	NPT party; AP in force Feb. 20, 2012; Pelindaba ratified 2012	Chinese state firms hold majority stakes in Rossing (CNNC 68.62%) and Husab (CGN 90%) [125]. Steps: uranium supply MOU; consider 123 or lesser cooperation framework [verify need]; Part 810 pathway
DRC	Historic uranium source; critical minerals partner	None	Specific authorization required	Not embargoed/restricted	NPT party; AP in force Apr. 9, 2003; Pelindaba ratified 2022	OFAC Part 547 sanctions program; U.S.-DRC Strategic Partnership Agreement (Dec. 2025) covers critical minerals [126]. Steps: responsible-sourcing and sanctions due diligence; capacity building

Country	Role	123 agreement	Part 810	Part 110	NPT / IAEA AP / Pelindaba	Gaps and needed steps
South Africa	Regional industrial and capital-market core; operates Koeberg; Meridian listing venue [127]	Expired Dec. 4, 2022; replacement negotiated at technical level but not signed as of latest public reports (Feb. 2025) [128, 129, 130]	Generally authorized (App. A)	NSG member; not embargoed/restricted	NPT party (acceded 1991); AP in force Sept. 13, 2002; Pelindaba ratified 1998	Conclude a new 123 agreement as the priority step; with no agreement in force, significant transfers of material and equipment cannot proceed [129]
Australia	Major uranium resource; ally	In force Dec. 22, 2010 to Dec. 22, 2040, then rolling 5-year extensions [110, 131]	Generally authorized (App. A)	NSG member	NPT; AP in force	Steps: confirm consent terms for Australian-obligated material in U.S. recycling [verify]
Canada	Second-largest producer (2024) [124]	In force (to 2030, rolling)	Generally authorized	NSG member	NPT; AP in force	Steps: same consent review; joint supply programs

Country	Role	123 agreement	Part 810	Part 110	NPT / IAEA AP / Pelindaba	Gaps and needed steps
Kazakhstan	Largest producer (~39% of world output, 2024) [124]; hosts IAEA LEU Bank [132]	In force Nov. 5, 1999, to Nov. 5, 2029 [110]	Generally authorized	NSG member	NPT; AP in force May 9, 2007	Renewal before Nov. 2029; about half of Western-bound shipments still transit Russia (2025; 48% used the Trans-Caspian route) [133]
Uzbekistan	Uranium producer	None	Specific authorization required	Not embargoed/restricted	NPT; AP in force Dec. 21, 1998	Critical-minerals framework signed Feb. 2026 [134]; consider nuclear cooperation framework
Niger	Uranium producer	None	Specific authorization required	Not embargoed/restricted	NPT; AP in force May 2, 2007; Pelindaba ratified 2016 (deposited 2017)	Political instability; Somair nationalization (2025) and ICSID arbitration [135, 136]. Steps: monitor; no near-term program role
France	Recycling leader (La Hague, Melox)	Via Euratom (1996, rolling)	Generally authorized	NSG member	NPT (NWS)	Steps: technology partnership; Part 810 specific authorization for any reprocessing assistance
United Kingdom	Fuel-cycle and safeguards partner	In force 2020-2050	Generally authorized	NSG member	NPT (NWS)	Steps: cooperation on plutonium disposition lessons
Japan	Recycling program (Rokkasho)	In force (1988, until terminated)	Generally authorized	NSG member	NPT; AP in force	Steps: plutonium balance coordination; joint standards

Country	Role	123 agreement	Part 810	Part 110	NPT / IAEA AP / Pelindaba	Gaps and needed steps
Republic of Korea	Advanced reactor and fuel partner	In force 2015-2040	Generally authorized	NSG member	NPT; AP in force	Steps: consultations under the agreement; policy coherence on domestic recycling

Table 12 has four practical implications. First, cooperation with Australia, Canada, Kazakhstan, and the allied fuel-cycle states is already broadly enabled. The remaining tasks are consent reviews, renewals, and specific authorizations for sensitive activities. Second, South Africa is generally authorized under Part 810 but has had no 123 agreement in force since December 2022 [128], which limits significant transfers. Third, Namibia, the DRC, Uzbekistan, and Niger are not Part 810 generally authorized destinations and have no 123 agreement. Any U.S. technical assistance therefore requires specific authorization. Fourth, uranium purchases from these suppliers are largely commercial transactions in natural uranium. They are governed by NRC import rules, sanctions law, and responsible-sourcing standards rather than by 123 agreements [verify for each transaction type].

11.3 Instruments for lawful trade and investment

- **New or amended 123 agreements.** Negotiated by State with DOE/NNSA and NRC consultation and submitted to Congress for review under AEA Sec. 123 [76, 110]. Prioritize renewal with Kazakhstan before 2029 and consider frameworks with Namibia and Uzbekistan where significant transfers are contemplated.
- **Part 810 streamlining for allied cooperative partners.** DOE could add qualifying partners to Appendix A for non-sensitive activities. It could also issue program-level specific authorizations for defined cooperative activities, while keeping case-by-case review for all reprocessing and plutonium-fuel assistance as Sec. 810.7 requires [116].
- **Updated country lists.** The NSG member list in 10 CFR 110.30 was last amended in 2014 [103]; NRC should keep such lists current [verify].
- **Fuel leasing and take-back.** Fuel-service offers, under which partners lease fuel and return spent fuel, reduce incentives for national sensitive facilities. They require host-state consent, transport agreements, and domestic legal authority to accept foreign spent fuel [verify current authority].
- **Multinational fuel assurance.** The IAEA LEU Bank in Kazakhstan, operational since October 2019 [132], is a model for assurance-of-supply mechanisms that complement a recycling cooperative.
- **Critical-minerals diplomacy.** FORGE, launched in February 2026 as successor to the Minerals Security Partnership [134], provides a venue to include uranium supply security and responsible sourcing.

- **Finance.** DFC lifted its prohibition on nuclear financing in 2020 [137]. EXIM and DFC can support mining, conversion, and fuel-service projects subject to nonproliferation and environmental review [verify program terms].
- **Investment and trade agreements.** Bilateral investment treaties and critical-minerals frameworks can protect investors against expropriation risk of the kind seen in Niger [135, 136].
- **Responsible for sourcing and sanctions.** For the DRC, OFAC screening, supply-chain due diligence, and transparency are essential. U.S. conflict-minerals reporting under Dodd-Frank Sec. 1502 covers tin, tantalum, tungsten, and gold, not uranium, so program standards should exceed legal minimums [verify].

11.4 Competition context

China's state nuclear companies hold majority stakes in Namibia's two largest operating mines and interests in others [125]. Russia remains a major fuel-cycle supplier subject to new U.S. import restrictions [121]. A U.S.-led cooperative offers partners an alternative built on transparency, safeguards, and fair commercial terms. That alternative pairs U.S. offtake and finance with capacity building.

11.5 An African anchor: the Meridian Economic and Infrastructure Charter

Southern Africa offers a natural first regional anchor for the cooperative. The proposed Meridian Charter for a Southern Africa economic infrastructure holding company, published as a public discussion document [127], would constitute a publicly listed development corporation, Meridian Economic and Infrastructure Holdings. Under treaties with willing host states and leases from owners who opt in, it would build and administer "consented economic polygons" around ports, rail, power nodes, and industrial estates. The Charter names Namibia, Botswana, South Africa, Zimbabwe, and Mozambique as potential host states. It proposes Walvis Bay and existing South African industrial zones as candidate first polygons, with listing in Johannesburg and London.

The Charter's capital plan has three layers. Sponsors provide seed equity for treaty counsel, survey work, and the first polygon. A public listing follows only after two host treaties and one polygon with audited revenue exist. Infrastructure debt sits at subsidiaries and is secured on port, power, and lease revenue, non-recourse to the host [127]. The hardest step is the first: credible, contracted revenue that makes seed capital investable.

A U.S.-led closed fuel cycle program could help supply that anchor. If Namibia, South Africa, and the Democratic Republic of the Congo joined the cooperative as supplier and partner states, program-linked commitments could give the Meridian ecosystem bankable cash flows and counterparties. The DRC is not one of the Charter's five named host states, so its participation would extend the Charter's regional scope and would need its own treaty basis. Possible mechanisms, all policy-level and subject to the export-control and sanctions rules in Section 11.1:

- **Uranium supply agreements.** Long-term, safeguarded supply arrangements with Namibian producers and, where responsible-sourcing standards are met, DRC sources. These create

predictable export volumes through port and logistics infrastructure that Meridian subsidiaries could build and operate.

- **Offtake contracts.** Program participants or U.S. fuel-cycle companies could sign offtake contracts, which the Charter expressly permits the company to sign [127]. Contracted revenue of this kind is what infrastructure lenders and listing committees look for.
- **DFC- and EXIM-backed financing tied to program participation.** DFC lifted its nuclear-financing prohibition in 2020 [137]. DFC and EXIM support for port, power, and logistics subsidiaries could be conditioned on partner-state adherence to cooperative membership criteria (Section 11.6), linking development finance to nonproliferation commitments [verify program eligibility].
- **Lawful-trade pathway.** South Africa is a Part 810 generally authorized destination and an NSG member, but it has no 123 agreement in force. Namibia and the DRC require Part 810 specific authorization for most assistance and have no 123 agreements [103, 110, 116]. Concluding a new U.S.-South Africa 123 agreement, and considering frameworks with Namibia, would give the corridor a durable legal basis.

This pairing serves both sides. The cooperative gains secure, diversified, and transparent uranium supply in a region where Chinese state firms now hold majority stakes in Namibia's largest mines [125]. The Meridian ecosystem gains an anchor customer and finance partner whose terms are published, safeguarded, and commercially disciplined. The Charter is a discussion document, not an offer of securities [127], and the cooperative proposed here does not depend on its adoption.

11.6 Proposed framework: a Closed Fuel Cycle Cooperative

1. Membership criteria: NPT party in full compliance; comprehensive safeguards with Additional Protocol (or, for nuclear-weapon states, voluntary offers); CPPNM Amendment adherence; NSG Guidelines adherence; effective export controls.
2. Roles: suppliers (uranium and conversion), fuel-cycle states (recycling and fabrication), and users (reactor operators leasing fuel).
3. Commitments: no national enrichment or reprocessing for users that obtain fuel services; no separated-plutonium stockpiles; INFCIRC/549-style transparency [56].
4. Benefits: assured supply and take-back services, financing access, technology cooperation on non-sensitive activities, and capacity building.
5. Governance: a secretariat coordinated by State and DOE with IAEA observers, and periodic independent review.

Table 13. Phased diplomatic roadmap (proposed targets)

Phase	Window	Milestones
A. Foundations	2026-2027	Interagency review of Part 810 Appendix A and Part 110 lists; consent review of supplier agreements; cooperative principles published
B. Core partners	2027-2029	Charter with France, UK, Japan, Canada, Australia, ROK; Kazakhstan 123 renewal before 2029 expiry [110]
C. Supplier compacts	2028-2031	Uranium supply and responsible-sourcing compacts with Namibia, South Africa, Uzbekistan, and others; Southern Africa anchor linked to Meridian corridor financing [127]; new U.S.-South Africa 123 agreement; DFC/EXIM project pipeline; DRC sourcing standards
D. Fuel services	2031 onward	Leasing and take-back pilots linked to U.S. recycling capacity; cooperative membership expanded

12. An Advocacy Plan: Nonproliferation and Roadblock Removal

12.1 Proliferation-resistant principles (conceptual)

The single greatest political obstacle to recycling in the United States has been, and remains, proliferation risk. A credible advocacy plan must lead with the solution rather than defend against the objection. The following principles are offered at a policy level only:

- **No separated pure plutonium.** Adopt as national policy that federally supported or licensed commercial recycling will not produce a separated, pure plutonium product. ARPA-E's CURIE program and several developers emphasize approaches that keep materials grouped rather than separated [8].

- **Co-processing and grouped recovery.** Favor approaches in which reusable materials are recovered together, reducing their attractiveness for misuse.
- **Safeguards-by-design.** Integrate safeguards, material accountancy, and monitoring into facilities from the earliest design stage, consistent with IAEA guidance [25, 55].
- **IAEA safeguards eligibility.** Make commercial recycling facilities eligible for IAEA safeguards under the U.S. voluntary offer agreement, demonstrating the transparency the United States asks of others.
- **Colocation.** Where practical, co-locate recycling, fuel fabrication, and reactors to minimize transport of sensitive materials, as envisioned in the integrated fast-reactor concept and in DOE's federal-site approach [12, 13].
- **No new states with sensitive capabilities.** Pair domestic recycling with international fuel-service offers so that partners have no need to develop their own.

12.2 Coalition strategy

Table 14. Coalition partners and value propositions

Partner	Value proposition	Ask
Utilities and owners	Defined end to storage liability; potential asset recognition; improved credit profile	Champion Standard Contract reform; pilot deliveries
Advanced reactor developers	Domestic feedstock; fuel security	Offtake commitments; co-investment
Organized labor	Skilled construction and operations jobs; continuity for nuclear trades	Public support; workforce programs
Host states and communities	Economic development; consent and benefits; removal of stranded fuel	Consent-based siting partnerships [23]
State regulators (NARUC, NWSC)	Return on ratepayer NWF payments; reduced customer cost	Rate frameworks; NWF reform [17]
National laboratories	Mission continuity; technology demonstration	Technical support, independent review
Nonproliferation NGOs	Policy safeguards they can endorse or shape	Participate in design of nonproliferation rules
Allies (France, Japan, UK, Canada, Korea)	Shared standards; supply chain partnership	Joint safeguards and fuel-service frameworks

Engaging nonproliferation organizations early, and inviting them to help write the rules, is more effective than trying to outvote them later. Their credibility with Congress and the public is an asset to a well-designed program.

12.3 Messaging

- **"Finish the job we paid for."** Ratepayers have paid more than $21 billion into the NWF and taxpayers more than $12 billion in damages [6, 7]; recycling is part of finally delivering a back-end solution.
- **"Turn a liability into a resource."** Spent fuel retains most of its energy potential [8].
- **"Recycling complements a repository."** Be explicit that a repository is still needed; recycling reduces its burden.
- **"Safer by design."** Lead with no-pure-plutonium and safeguards-by-design commitments.
- **"Built in America, by Americans."** Domestic jobs, fuel security, and leadership in global nuclear commerce.

Messages should avoid overstated claims such as "recycling eliminates nuclear waste." Overclaiming invites rebuttal and erodes credibility with the policymakers and investors this campaign needs.

12.4 Phased pilot pathway and milestones

Table 15. Phased pathway and timeline (proposed; dates are targets, not commitments)

Phase	Target window	Key milestones	Gate to next phase
0. Policy foundation	2026-2027	DOE national policy under EO 14302 published [verify public release status]; NRC final rule [verify target date]; NWPA recycling-pathway bill introduced; FERC/PUC dialogue begun	Legal authority and licensing pathway in place
1. Pilot demonstration	2027-2031	Selections under DOE RFAs; pilot on a federal site with safeguards-by-design; independent cost review; first Standard Contract amendment pilot	Demonstrated safeguards performance and cost data
2. First commercial facility	2030-2036	Licensed first-of-a-kind commercial facility (e.g., Oklo's planned 2030 commissioning target [16]); utility delivery contracts; first accounting recognition under contract	Contract-based valuation; residual-waste pathway defined
3. Fleet integration	2035 onward	Recycled fuel supplying advanced reactors; regional facilities; repository design reflecting recycled waste forms	Periodic national review of program economics and security

Each phase gate should require an independent review, by a body such as the National Academies or GAO, of cost, safeguards, and waste performance. Gates convert a long-term vision into a sequence of decisions that Congress, investors, and the public can evaluate.

13. Conclusions and Recommendations

The United States is paying for a back-end system it does not have. Ratepayers funded a Nuclear Waste Fund now exceeding $51 billion; taxpayers have paid about $12.2 billion in damages with tens of billions more expected; and utilities carry spent fuel as an indefinite liability [6, 7]. The policy, regulatory, and industrial conditions for a different approach are now emerging faster than at any time since 1977.

A modern, proliferation-resistant closed fuel cycle would not, by itself, make every kilogram of spent fuel an asset. It would give spent fuel a defined, contractible destination, create the conditions under which some recovered material can be recognized as having value, and reduce the burden on a future repository. That is the shift from the liability column toward the asset column that improves the finances of existing plants and the investment case for new ones.

1. Congress should amend the NWPA to create a voluntary recycling pathway that satisfies DOE acceptance obligations and defines disposal of residual waste.
2. Congress and OMB should reform the Nuclear Waste Fund's budget treatment so that funds already collected can support an integrated back-end program.
3. DOE should complete the national fuel-cycle policy under EO 14302, execute its recycling RFAs with transparent milestones, and develop a Standard Contract amendment template.
4. The NRC should finalize a licensing pathway for recycling facilities that incorporates safeguards-by-design and security from the outset.
5. FERC, state commissions, and accounting standard-setters should clarify how recovered materials can be recognized once a contractual or market basis exists, using the existing net-salvage-value framework as a starting point.
6. The Administration should adopt no-pure-plutonium, safeguards-by-design, IAEA-eligibility, and colocation as binding principles for federally supported recycling.
7. Congress, the NRC, DOE, EPA, and the White House should act on the citation-level changes in Section 10, starting with the NWPA, the Standard Contract, PDD-13 policy, and the NRC Part 70 recycling rule.
8. State, DOE, the NRC, and Commerce should launch a Global Closed Fuel Cycle Cooperative within the U.S. nuclear export control regime: review Part 810 Appendix A and Part 110 lists, renew the U.S.-Kazakhstan 123 agreement before 2029, and use specific authorizations for all sensitive activities (Section 11).
9. Industry, labor, states, national laboratories, and nonproliferation organizations should form a standing coalition to support a phased program with independent review at each gate.

About the Author

Keith J Maxwell is a consultant working in engineering, naval architecture, project management, and nuclear and energy topics. He has USN, US Government and commercial nuclear design, licensing, operations, maintenance, and modification experience.

Disclaimer: This paper presents the author's policy views for discussion. It is not legal, accounting, tax, or investment advice. Illustrative economics are hypothetical and are not forecasts.

References

[1] Code of Federal Regulations, 10 CFR Part 810, Assistance to Foreign Atomic Energy Activities (DOE). https://www.ecfr.gov/current/title-10/chapter-III/part-810

[2] Code of Federal Regulations, 10 CFR Part 110, Export and Import of Nuclear Equipment and Material (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-110

[3] U.S. Department of Energy, Office of Nuclear Energy / Idaho National Laboratory (Banerjee, K., and Gadey, H.). Spent Nuclear Fuel and Reprocessing Waste Inventory, INL/RPT-25-88353 Rev. 2 (inventory as of Dec. 31, 2024; published Jan. 2026). https://doi.org/10.2172/3015718

[4] U.S. DOE Office of Nuclear Energy. "5 Fast Facts about Spent Nuclear Fuel." https://www.energy.gov/ne/articles/5-fast-facts-about-spent-nuclear-fuel

[5] Code of Federal Regulations, 10 CFR Part 961, Standard Contract for Disposal of Spent Nuclear Fuel and/or High-Level Radioactive Waste. https://www.ecfr.gov/current/title-10/chapter-III/part-961

[6] U.S. Department of the Treasury, Bureau of the Fiscal Service. Financial Report of the U.S. Government FY2025, Note 21: Contingencies. https://fiscal.treasury.gov/system/files/2026-04/2025-note-21-financial-statements.pdf

[7] U.S. DOE. Nuclear Waste Fund (NWF) Annual Financial Report Summary, FY2025 and Cumulative. https://www.energy.gov/documents/fy-2025-nwf-annual-financial-report-summary

[8] ARPA-E. CURIE (Converting UNF Radioisotopes Into Energy) program page. https://arpa-e.energy.gov/programs-and-initiatives/view-all-programs/curie

[9] Orano. Annual Activity Report 2024 (Recycling section). https://www.orano.group/finance-raa2024/en/40/

[10] French Ministry for Ecological Transition. Plan national de gestion des matières et des déchets radioactifs (PNGMDR): La gestion des matières radioactives. https://dechets-radioactifs.ecologie.gouv.fr/la-gestion-des-matieres-radioactives-32

[11] The White House. Executive Order 14302, "Reinvigorating the Nuclear Industrial Base" (May 23, 2025). https://www.whitehouse.gov/presidential-actions/2025/05/reinvigorating-the-nuclear-industrial-base/

[12] Federal Register. Executive Order 14299, "Deploying Advanced Nuclear Reactor Technologies for National Security" (May 23, 2025; published May 29, 2025). https://www.federalregister.gov/documents/2025/05/29/2025-09796/deploying-advanced-nuclear-reactor-technologies-for-national-security

[13] U.S. DOE Office of Environmental Management. "Department of Energy Seeks to Partner with Private Sector for Used Nuclear Fuel Recycling" (Apr. 22, 2026). https://www.energy.gov/em/articles/department-energy-seeks-partner-private-sector-used-nuclear-fuel-recycling

[14] U.S. DOE Office of Nuclear Energy. Advanced Nuclear Fuel Recycling Program. https://www.energy.gov/ne/advanced-nuclear-fuel-recycling-program

[15] U.S. Nuclear Regulatory Commission. "Modernizing Materials Licensing," proposed rule, Federal Register (June 24, 2026). https://www.federalregister.gov/documents/2026/06/24/2026-12702/modernizing-materials-licensing

[16] Oklo Inc. "Oklo Announces Fuel Recycling Facility as First Phase of up to $1.68 Billion Advanced Fuel Center in Tennessee" (Sept. 4, 2025). https://oklo.com/newsroom/oklo-announces-fuel-recycling-facility-as-first-phase-of-up-to-1-68-billion-advanced-fuel-center

[17] Nuclear Waste Strategy Coalition, written testimony to the U.S. House Energy and Commerce Committee, "Nuclear Spent Fuel Policy" hearing (Sept. 15, 2026). https://www.congress.gov/119/meeting/house/119550/witnesses/HMTG-119-IF03-Wstate-WhiteG-20260915.pdf

[18] Written testimony (McMurrian) before the Senate Environment and Public Works Committee, "Hearing to Examine Updating America's Nuclear Waste Management Policy" (Sept. 23, 2026). https://www.epw.senate.gov/public/_cache/files/a/5/a5eac4ee-d904-422b-b64e-51049ae72e44/A76890188E8BC9950B242C770344C9F4598294EDC1811180C4BDB8D0254612E9.09-23-2026-mcmurrian-testimony.pdf

[19] MIT Energy Initiative. The Future of the Nuclear Fuel Cycle: An Interdisciplinary MIT Study (2011). https://energy.mit.edu/publication/future-nuclear-fuel-cycle/

[20] Bunn, M., Fetter, S., Holdren, J., and van der Zwaan, B. The Economics of Reprocessing vs. Direct Disposal of Spent Nuclear Fuel, Harvard Belfer Center (2003). https://www.belfercenter.org/publication/economics-reprocessing-vs-direct-disposal-spent-nuclear-fuel

[21] U.S. Government Accountability Office. Commercial Spent Nuclear Fuel: Congressional Action Needed to Break Impasse and Develop a Permanent Disposal Solution, GAO-21-603 (Sept. 2021). https://www.gao.gov/products/gao-21-603

[22] Blue Ribbon Commission on America's Nuclear Future. Report to the Secretary of Energy (Jan. 2012). https://www.energy.gov/ne/articles/blue-ribbon-commission-americas-nuclear-future-report-secretary-energy

[23] Bipartisan Policy Center Nuclear Waste Council. Moving Forward with Consent-Based Siting for Nuclear Waste Facilities (2019). https://bipartisanpolicy.org/wp-content/uploads/2019/03/Nuclear-Consent-Based-Siting.pdf

[24] Code of Federal Regulations, 18 CFR Part 101, FERC Uniform System of Accounts (Accounts 120.1-120.6, 230, 518). https://www.ecfr.gov/current/title-18/chapter-I/subchapter-C/part-101

[25] International Atomic Energy Agency. Safeguards by Design. https://www.iaea.org/topics/assistance-for-states/safeguards-by-design

[26] U.S. Court of Appeals for the D.C. Circuit. National Association of Regulatory Utility Commissioners v. U.S. Department of Energy, No. 11-1066 (Nov. 19, 2013). https://media.cadc.uscourts.gov/opinions/docs/2013/11/11-1066-1466796.pdf

[27] World Nuclear News. "Zero day for US nuclear waste fee" (May 16, 2014). https://www.world-nuclear-news.org/articles/zero-day-for-us-nuclear-waste-fee

[28] World Nuclear Association. Processing of Used Nuclear Fuel. https://world-nuclear.org/information-library/nuclear-fuel-cycle/fuel-recycling/processing-of-used-nuclear-fuel

[29] OECD Nuclear Energy Agency. The Economics of the Back End of the Nuclear Fuel Cycle, NEA No. 7061 (2013). https://www.oecd-nea.org/upload/docs/application/pdf/2019-12/7061-ebenfc.pdf

[30] U.S. DOE Office of Inspector General. The Department of Energy Nuclear Waste Fund's Fiscal Year 2025 Financial Statement Audit, DOE-OIG-26-02 (Nov. 2025). https://www.energy.gov/ig/articles/financial-statement-audit-doe-oig-26-02

[31] ASNR (French Nuclear Safety and Radiation Protection Authority). Opinion on the management of radioactive materials and their recoverability, in preparation of the 6th PNGMDR (2026). https://www.asnr.fr/actualites/dans-le-cadre-de-la-preparation-du-6e-pngmdr-lasnr-rend-son-avis-sur-la-gestion-des

[32] International Panel on Fissile Materials. Marignac, Y., and Schneider, M., et al. Spent Nuclear Fuel Reprocessing in France, Research Report No. 4 (2008). https://fissilematerials.org/library/rr04.pdf

[33] PricewaterhouseCoopers Audit and KPMG SA. Électricité de France: Rapport des commissaires aux comptes sur les comptes consolidés, exercice clos le 31 décembre 2025 (Feb. 19, 2026). https://www.edf.fr/sites/groupe/files/2026-02/resultats-annuels-2025-rapport-cac.pdf

[34] U.S. NRC. SECY-21-0026, Discontinuation of Rulemaking - Spent Fuel Reprocessing (2021). https://www.nrc.gov/docs/ML2030/ML20301A388.pdf

[35] Orano. "All about nuclear and radioactive waste in France." https://www.orano.group/en/unpacking-nuclear/all-about-radioactive-waste-in-france

[36] Orano. "Understanding nuclear fuel recycling." https://www.orano.group/en/unpacking-nuclear/all-about-used-fuel-processing-and-recycling

[37] Orano. Response to WISE-Paris (question Q1a), French national public debate on the PNGMDR (Commission particuliere du debat public). https://pngmdr.debatpublic.fr/images/contenu/documentation/clarification-controverses/Q1a_Wise_2_Orano.pdf

[38] EASAC / European Commission Joint Research Centre. Management of Spent Nuclear Fuel and its Waste, JRC Science and Policy Report (2014), summarizing OECD-NEA analyses. https://publications.jrc.ec.europa.eu/repository/bitstream/JRC84826/jrc-report-anagement-spent-fuel-and-waste.pdf

[39] IAEA. Implications of Partitioning and Transmutation in Radioactive Waste Management, Technical Reports Series No. 435 (2004). https://www-pub.iaea.org/MTCD/Publications/PDF/TRS435_web.pdf

[40] OECD-NEA. "Challenges and potential benefits of partitioning and transmutation (P&T)," NEA News Vol. 24 No. 1 (2006). https://oecd-nea.org/pub/newsletter/2006/NEA_News_Vol_24_No_1_p_t.pdf

[41] U.S. NRC. "Requirements for the Storage of Spent Nuclear Fuel," Federal Register (Jan. 23, 2020). https://www.federalregister.gov/documents/2020/01/23/2020-01026/requirements-for-the-storage-of-spent-nuclear-fuel

[42] U.S. NRC. "Continued Storage of Spent Nuclear Fuel," final rule, Federal Register (Sept. 19, 2014), codifying NUREG-2157 generic determinations in 10 CFR 51.23. https://www.federalregister.gov/documents/2014/09/19/2014-22215/continued-storage-of-spent-nuclear-fuel

[43] Hamilton, L., and Scowcroft, B. Statement of the Co-Chairmen, Blue Ribbon Commission on America's Nuclear Future, before the Senate Committee on Energy and Natural Resources (Feb. 2, 2012). https://www.energy.senate.gov/wp-content/uploads/meetings/73134538-E261-4B1D-B2A9-A82817EAE75C/Hamilton%20and%20Scowcroft%20testimony%2C%2002.02.12.pdf

[44] OSPAR Commission. Discharges of Radioactive Substances from the Nuclear Sector, 2024 assessment. https://oap.ospar.org/en/ospar-assessments/committee-assessments/radioactive-substances/discharges-nuclear-installations/nuc-2024/

[45] Code of Federal Regulations, 40 CFR Part 190, Environmental Radiation Protection Standards for Nuclear Power Operations (EPA). https://www.ecfr.gov/current/title-40/chapter-I/subchapter-F/part-190

[46] Lyman, E., and Feiveson, H. "The Proliferation Risks of Plutonium Mines," Science & Global Security 7 (1998). https://doi.org/10.1080/08929889808426449

[47] Code of Federal Regulations, 10 CFR Part 73, Physical Protection of Plants and Materials (NRC), incl. Sec. 73.46. https://www.ecfr.gov/current/title-10/chapter-I/part-73

[48] Code of Federal Regulations, 10 CFR Part 74, Material Control and Accounting of Special Nuclear Material (NRC), incl. Subpart E. https://www.ecfr.gov/current/title-10/chapter-I/part-74

[49] Union of Concerned Scientists. "Nuclear Reprocessing: Dangerous, Dirty, and Expensive." https://www.ucs.org/resources/nuclear-reprocessing-dangerous-dirty-and-expensive

[50] UK Parliament. Written Ministerial Statement HCWS388, "Plutonium Disposition Strategy" (Jan. 24, 2025). https://www.theyworkforyou.com/wms/?id=2025-01-24.hcws388.h

[51] Arms Control Association. "Russia Suspends Plutonium Agreement," Arms Control Today (Oct. 2016). https://www.armscontrol.org/act/2016-10/news/russia-suspends-plutonium-agreement

[52] U.S. NRC. Mixed Oxide Fuel Fabrication Facility Licensing. https://www.nrc.gov/materials/fuel-cycle-fac/mox/licensing

[53] Reuters. "Trump administration kills contract for plutonium-to-fuel plant" (Oct. 2018). https://www.reuters.com/article/us-usa-plutonium-mox/trump-administration-kills-contract-for-plutonium-to-fuel-plant-idUSKCN1MM2N0/

[54] U.S. DOE/NNSA. Surplus Plutonium Disposition Program Environmental Impact Statement fact sheet. https://www.energy.gov/nnsa/articles/surplus-plutonium-disposition-program-sdsp-environmental-impact-statement-fact-sheet

[55] IAEA. International Safeguards in Nuclear Facility Design and Construction, IAEA Nuclear Energy Series (2013). https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1600_web.pdf

[56] IAEA. INFCIRC/549, Communication Received from Certain Member States Concerning Their Policies Regarding the Management of Plutonium (Guidelines for the Management of Plutonium). https://www.iaea.org/sites/default/files/infcirc549.pdf

[57] Congressional Budget Office. Testimony on the Federal Government's Responsibilities and Liabilities Under the Nuclear Waste Policy Act (Dec. 3, 2015). https://www.cbo.gov/publication/51035

[58] U.S. DOE. Analysis of the Total System Life Cycle Cost of the Civilian Radioactive Waste Management Program, Fiscal Year 2007, DOE/RW-0591 (July 2008). https://www.energy.gov/gc/articles/fy-2007-total-system-life-cycle-cost-pub-2008

[59] U.S. DOE. Fiscal Year 2007 Civilian Radioactive Waste Management Fee Adequacy Assessment Report (2008). https://www.energy.gov/gc/articles/fy-2007-fee-adequacy-pub-2008-0

[60] NYSERDA. West Valley: Fuel Reprocessing History. https://www.nyserda.ny.gov/All-Programs/West-Valley/Fuel-Reprocessing-History

[61] U.S. NRC. NUREG/CR-7232, Review of Spent Fuel Reprocessing and Associated Accident Phenomena (2017) (historical background sections only). https://www.nrc.gov/docs/ML1704/ML17045A577.pdf

[62] U.S. NRC. Notice of issuance of renewed license SNM-2500, GE Hitachi Morris Operation ISFSI (renewed Nov. 22, 2022; expires May 31, 2042), 87 FR 74177 (Dec. 2, 2022). https://www.govinfo.gov/content/pkg/FR-2022-12-02/html/2022-26230.htm

[63] Congressional Research Service. Nuclear Fuel Reprocessing: U.S. Policy Development, RS22542. https://www.congress.gov/crs_external_products/RS/PDF/RS22542/RS22542.5.pdf

[64] Breeder Reactor Corporation. The Clinch River Breeder Reactor Plant Project Final Report (NRC ADAMS ML18064A893). https://www.nrc.gov/docs/ML1806/ML18064A893.pdf

[65] Westfall, C. "Vision and Reality: The EBR-II Story," Nuclear News, American Nuclear Society (Feb. 2004). https://www2.ans.org/pubs/magazines/nn/docs/2004-2-2.pdf

[66] U.S. GAO. Global Nuclear Energy Partnership: DOE Should Reassess Its Approach to Designing and Building Spent Nuclear Fuel Recycling Facilities, GAO-08-483 (2008). https://www.gao.gov/products/gao-08-483

[67] Federal Register. Notice of Cancellation of the Global Nuclear Energy Partnership (GNEP) Programmatic Environmental Impact Statement (June 29, 2009). https://www.federalregister.gov/documents/2009/06/29/E9-15328/notice-of-cancellation-of-the-global-nuclear-energy-partnership-gnep-programmatic-environmental

[68] World Nuclear News. "US DOE awards funds for nuclear recycling projects" (Oct. 2022). https://world-nuclear-news.org/articles/us-doe-awards-funds-for-nuclear-recycling-projects

[69] Japan Nuclear Fuel Limited. "Change in Schedule for Completion of Rokkasho Reprocessing Plant and MOX Fabrication Plant" (Aug. 29, 2024). https://www.jnfl.co.jp/en/release/president-talk/2024/202408.html

[70] Nikkei Asia. "Japan nuclear fuel reprocessing plant at risk of 28th delay" (2026). https://asia.nikkei.com/business/energy/japan-nuclear-fuel-reprocessing-plant-at-risk-of-28th-delay

[71] UK Government / Nuclear Decommissioning Authority. "Job done: Sellafield plant safely completes its mission" (Magnox reprocessing ends, 2022). https://www.gov.uk/government/news/job-done-sellafield-plant-safely-completes-its-mission

[72] World Nuclear Association. Russia's Nuclear Fuel Cycle. https://world-nuclear.org/information-library/country-profiles/countries-o-s/russia-nuclear-fuel-cycle

[73] Government of India, Department of Atomic Energy. "Prototype Fast Breeder Reactor at Kalpakkam, Tamil Nadu attains First Criticality" (Apr. 7, 2026). https://dae.gov.in/prototype-fast-breeder-reactor-at-kalpakkam-tamil-nadu-attains-first-criticality/

[74] U.S. Code, Title 42, Chapter 108, Nuclear Waste Policy (Nuclear Waste Policy Act of 1982, as amended), 42 U.S.C. 10101 et seq. https://uscode.house.gov/view.xhtml?path=/prelim@title42/chapter108&edition=prelim

[75] Code of Federal Regulations, 10 CFR Part 960, General Guidelines for the Preliminary Screening of Potential Sites for a Nuclear Waste Repository (DOE). https://www.ecfr.gov/current/title-10/chapter-III/part-960

[76] U.S. Code, Title 42, Chapter 23, Development and Control of Atomic Energy (Atomic Energy Act of 1954, as amended), 42 U.S.C. 2011 et seq. https://uscode.house.gov/view.xhtml?path=/prelim@title42/chapter23&edition=prelim

[77] U.S. GAO. B-230201, review of the proposed U.S.-Japan Agreement for Cooperation (Feb. 29, 1988). https://www.gao.gov/products/b-230201

[78] Code of Federal Regulations, 10 CFR Part 140, Financial Protection Requirements and Indemnity Agreements (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-140

[79] U.S. NRC. Backgrounder/fact sheet on nuclear insurance and the Price-Anderson Act. https://www.nrc.gov/docs/ML1027/ML102720179.pdf

[80] U.S. Code, Title 22, Chapter 58, Nuclear Non-Proliferation (Nuclear Non-Proliferation Act of 1978, Pub. L. 95-242), 22 U.S.C. 3201 et seq. https://uscode.house.gov/view.xhtml?path=/prelim@title22/chapter58&edition=prelim

[81] U.S. Code, Title 42, Chapter 55, National Environmental Policy (NEPA), 42 U.S.C. 4321 et seq. https://uscode.house.gov/view.xhtml?path=/prelim@title42/chapter55&edition=prelim

[82] Council on Environmental Quality. "Removal of National Environmental Policy Act Implementing Regulations," interim final rule, Federal Register (Feb. 25, 2025; effective Apr. 11, 2025). https://www.federalregister.gov/documents/2025/02/25/2025-03014/removal-of-national-environmental-policy-act-implementing-regulations

[83] U.S. DOE. "Revision of National Environmental Policy Act Implementing Procedures," interim final rule, Federal Register (July 3, 2025). https://www.federalregister.gov/documents/2025/07/03/2025-12383/revision-of-national-environmental-policy-act-implementing-procedures

[84] Public Law 118-67 (July 9, 2024), including the ADVANCE Act of 2024. https://www.congress.gov/118/plaws/publ67/PLAW-118publ67.pdf

[85] Code of Federal Regulations, 40 CFR Part 191, Environmental Radiation Protection Standards for Management and Disposal of Spent Nuclear Fuel, High-Level and Transuranic Radioactive Wastes (EPA). https://www.ecfr.gov/current/title-40/chapter-I/subchapter-F/part-191

[86] Code of Federal Regulations, 40 CFR Part 197, Public Health and Environmental Radiation Protection Standards for Yucca Mountain, Nevada (EPA). https://www.ecfr.gov/current/title-40/chapter-I/subchapter-F/part-197

[87] The White House, Office of the Press Secretary. Fact Sheet: Nonproliferation and Export Control Policy (Sept. 27, 1993) (PDD-13 policy statement). https://csps.aerospace.org/sites/default/files/2021-08/Non-Proliferation%20%26%20Export%20Control%20Policy%20Sep93.pdf

[88] The White House. Executive Order 14300, "Ordering the Reform of the Nuclear Regulatory Commission" (May 23, 2025). https://www.whitehouse.gov/presidential-actions/2025/05/ordering-the-reform-of-the-nuclear-regulatory-commission/

[89] California Public Resources Code Sec. 25524.2 (nuclear powerplant certification conditioned on approved high-level waste disposal technology). https://leginfo.legislature.ca.gov/faces/codes_displaySection.xhtml?lawCode=PRC&sectionNum=25524.2

[90] U.S. NRC. "Risk-Informed, Technology-Inclusive Regulatory Framework for Advanced Reactors," final rule adding 10 CFR Part 53, 91 FR 15696 (Mar. 30, 2026; effective Apr. 29, 2026). https://www.federalregister.gov/documents/2026/03/30/2026-06048/risk-informed-technology-inclusive-regulatory-framework-for-advanced-reactors

[91] Code of Federal Regulations, 10 CFR Part 20, Standards for Protection Against Radiation (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-20

[92] Code of Federal Regulations, 10 CFR Part 50, Domestic Licensing of Production and Utilization Facilities (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-50

[93] Code of Federal Regulations, 10 CFR Part 51, Environmental Protection Regulations for Domestic Licensing and Related Regulatory Functions (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-51

[94] Code of Federal Regulations, 10 CFR Part 52, Licenses, Certifications, and Approvals for Nuclear Power Plants (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-52

[95] Code of Federal Regulations, 10 CFR Part 60, Disposal of High-Level Radioactive Wastes in Geologic Repositories (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-60

[96] Code of Federal Regulations, 10 CFR Part 61, Licensing Requirements for Land Disposal of Radioactive Waste (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-61

[97] Code of Federal Regulations, 10 CFR Part 63, Disposal of High-Level Radioactive Wastes in a Geologic Repository at Yucca Mountain, Nevada (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-63

[98] Code of Federal Regulations, 10 CFR Part 70, Domestic Licensing of Special Nuclear Material (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-70

[99] Code of Federal Regulations, 10 CFR Part 71, Packaging and Transportation of Radioactive Material (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-71

[100] Code of Federal Regulations, 10 CFR Part 72, Licensing Requirements for the Independent Storage of Spent Nuclear Fuel, High-Level Radioactive Waste, and Reactor-Related Greater Than Class C Waste (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-72

[101] Code of Federal Regulations, 10 CFR Part 75, Safeguards on Nuclear Material: Implementation of Safeguards Agreements Between the United States and the IAEA (NRC). https://www.ecfr.gov/current/title-10/chapter-I/part-75

[102] IAEA. INFCIRC/288, Agreement between the United States of America and the IAEA for the Application of Safeguards in the United States (and INFCIRC/288/Add.1, Additional Protocol). https://www.iaea.org/publications/documents/infcircs/text-agreement-18-november-1977-between-united-states-america-and-agency-application-safeguards-united-states-america

[103] Code of Federal Regulations, 10 CFR Part 110, Secs. 110.20-110.29 (general licenses; embargoed and restricted destinations) and 110.30 (NSG members). https://www.ecfr.gov/current/title-10/chapter-I/part-110

[104] Treaty on the Non-Proliferation of Nuclear Weapons, IAEA INFCIRC/140 (1970). https://www.iaea.org/sites/default/files/publications/documents/infcircs/1970/infcirc140.pdf

[105] U.S. NRC. Additional Protocol to the U.S.-IAEA Safeguards Agreement (entered into force Jan. 6, 2009). https://www.nrc.gov/materials/fuel-cycle-fac/intl-safeguards/addtl-protocol-us-iaea-safeguards-agrmnt

[106] IAEA. Convention on the Physical Protection of Nuclear Material (CPPNM) and its 2005 Amendment (in force May 8, 2016). https://www.iaea.org/publications/documents/conventions/convention-physical-protection-nuclear-material-and-its-amendment

[107] IAEA Office of Legal Affairs. United States: status of multilateral conventions (CPPNM, Joint Convention, CSC). https://ola.iaea.org/Applications/FactSheets/Country/Detail?code=US

[108] IAEA. INFCIRC/546, Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management (Article 3). https://www.iaea.org/sites/default/files/infcirc546.pdf

[109] Agreement for Cooperation in the Peaceful Uses of Nuclear Energy between Euratom and the United States (1996), EUR-Lex text. https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX%3A21996A0520%2801%29

[110] U.S. DOE/NNSA. 123 Agreements for Peaceful Cooperation (list and terms). https://www.energy.gov/nnsa/123-agreements-peaceful-cooperation

[111] Agreement for Cooperation between the Republic of Korea and the United States Concerning Peaceful Uses of Nuclear Energy (2015), IPFM library copy. https://fissilematerials.org/library/kr123.pdf

[112] Nuclear Suppliers Group. Guidelines for Nuclear Transfers, IAEA INFCIRC/254/Rev.11/Part 1 (paragraphs 6-7). https://www.vertic.org/media/assets/nim_docs/NSG%20and%20ZC/infcirc254r11p1.pdf

[113] Zangger Committee. Communications on exports of nuclear material and equipment (trigger list), IAEA INFCIRC/209 series. https://www.zanggercommittee.org/

[114] IAEA. Convention on Supplementary Compensation for Nuclear Damage: status (in force Apr. 15, 2015). https://www.iaea.org/sites/default/files/22/06/supcomp_status.pdf

[115] U.S. Department of State. Treaties and Other International Acts Series 15-415: Convention on Supplementary Compensation for Nuclear Damage (text incl. Annex). https://www.state.gov/wp-content/uploads/2019/02/15-415-Multilateral-Atomic-Energy-Compensation.pdf

[116] Code of Federal Regulations, 10 CFR Part 810, Sec. 810.6-810.8 and Appendix A (Generally Authorized Destinations), as amended through 91 FR 36073 (June 16, 2026). https://www.ecfr.gov/current/title-10/chapter-III/part-810

[117] Code of Federal Regulations, 15 CFR Part 742, Control Policy: CCL Based Controls (incl. Sec. 742.3, Nuclear Nonproliferation) (Commerce/BIS). https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-742

[118] Code of Federal Regulations, 15 CFR Part 744, Control Policy: End-User and End-Use Based (incl. Sec. 744.2 nuclear end-uses; Supplement No. 4, Entity List) (Commerce/BIS). https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-744

[119] Code of Federal Regulations, 22 CFR Parts 120-130, International Traffic in Arms Regulations (State/DDTC). https://www.ecfr.gov/current/title-22/chapter-I/subchapter-M

[120] Code of Federal Regulations, 31 CFR Chapter V, Office of Foreign Assets Control (Treasury), incl. Part 547 (Democratic Republic of the Congo Sanctions Regulations). https://www.ecfr.gov/current/title-31/subtitle-B/chapter-V

[121] Public Law 118-62, Prohibiting Russian Uranium Imports Act (2024). https://www.govinfo.gov/content/pkg/PLAW-118publ62/html/PLAW-118publ62.htm

[122] IAEA. Status of the Additional Protocol (list of States). https://www.iaea.org/sites/default/files/AP_status_list.pdf

[123] African Union. List of countries which have signed, ratified/acceded to the African Nuclear-Weapon-Free Zone Treaty (Treaty of Pelindaba). https://www.au.int/sites/default/files/treaties/37288-sl-THE_AFRICAN_NUCLEAR-WEAPON-FREE_ZONE_TREATY_THE_TREATY_OF_PELINDABA_2.pdf

[124] World Nuclear Association. World Uranium Mining Production. https://world-nuclear.org/information-library/nuclear-fuel-cycle/mining-of-uranium/world-uranium-mining-production

[125] Mining & Energy (Namibia). "Chinese firms increase dominance in Namibia's uranium sector" (May 15, 2026). https://miningandenergy.com.na/chinese-firms-increase-dominance-in-namibias-uranium-sector/

[126] U.S. Department of State. Strategic Partnership Agreement Between the United States and the Democratic Republic of the Congo (Dec. 2025). https://www.state.gov/strategic-partnership-agreement-between-the-government-of-the-united-states-of-america-and-the-government-of-the-democratic-republic-of-the-congo

[127] Maxwell, Keith J. Meridian Economic and Infrastructure Charter: A Public Proposal. Discussion document, October 2026. https://maxwellassociates.net/stories/pdfs/meridian-economic-infrastructure-charter.pdf

[128] U.S. DOE. Agreement for Cooperation between the United States and the Republic of South Africa Concerning Peaceful Uses of Nuclear Energy (status: expired Dec. 4, 2022). https://www.energy.gov/node/4812839

[129] South Africa Department of Mineral Resources and Energy. Media statement on expiry of the South Africa-USA Nuclear Cooperation Agreement (2023). https://www.gov.za/news/media-statements/mineral-resources-and-energy-expiry-south-africa-and-usa-nuclear-cooperation

[130] CNBC Africa (Reuters). "South Africa open to nuclear project bids from Russia or Iran, minister says" (Feb. 2025). https://www.cnbcafrica.com/2025/south-africa-open-to-nuclear-project-bids-from-russia-or-iran-minister-says

[131] U.S. Department of State. Treaties and Other International Acts Series 10-1222: Agreement for Cooperation with Australia Concerning Peaceful Uses of Nuclear Energy. https://www.state.gov/10-1222

[132] IAEA. IAEA Low Enriched Uranium (LEU) Bank (Ulba Metallurgical Plant, Kazakhstan). https://www.iaea.org/topics/iaea-low-enriched-uranium-bank

[133] Kazatomprom. FY2025 Operations and Trading Update / Operating and Financial Review (2026). https://www.kazatomprom.kz/storage/86/fy2025_ofr_eng.pdf

[134] U.S. Department of State. 2026 Critical Minerals Ministerial fact sheet (Feb. 4, 2026), announcing FORGE as successor to the Minerals Security Partnership. https://www.state.gov/releases/office-of-the-spokesman/2026/02/2026-critical-minerals-ministerial

[135] Orano. "Orano opposes nationalization plans of SOMAIR in Niger" (June 20, 2025). https://www.orano.group/en/news/news-group/2025/june/orano-opposes-nationalization-plans-of-somair-in-niger

[136] ICSID. Orano Mining SAS v. Republic of Niger, ICSID Case No. ARB/25/9 (case details). https://icsid.worldbank.org/cases/case-database/case-detail?CaseNo=ARB%2F25%2F9

[137] U.S. International Development Finance Corporation. "DFC Modernizes Nuclear Energy Policy" (2020). https://www.dfc.gov/media/press-releases/dfc-modernizes-nuclear-energy-policy